

HSPT Exam Simulator

Full-Length Practice Exam 08

Complete printable test with 298 multiple-choice questions, answer key, and detailed explanations.

Section	Questions	Time
Verbal Skills	60	20 minutes
Quantitative Skills	52	35 minutes
Reading Comprehension	62	35 minutes
Mathematics	64	45 minutes
Language	60	30 minutes

General Directions

Work through each section within the recommended time. Choose the best answer for every question. Mark one answer (A-D) for each item. Use the answer key and explanations only after completing the test.

Recommended total testing time: 165 minutes. Calculators should be used only when permitted by the rules followed for your HSPT preparation.

Section 1: Verbal Skills

60 questions | Recommended time: 20 minutes

1. Which word is closest in meaning to deliberate?

- A) accidental
- B) rapid
- C) careful
- D) ordinary

2. Which word is closest in meaning to obscure?

- A) hide
- B) repair
- C) celebrate
- D) measure

3. Which word is closest in meaning to novice?

- A) leader
- B) beginner
- C) critic
- D) expert

4. Which word is most nearly opposite in meaning to fragile?

- A) delicate
- B) smooth
- C) sturdy
- D) small

5. Which word is most nearly opposite in meaning to frequent?

- A) common
- B) regular
- C) expected
- D) rare

6. Which word is most nearly opposite in meaning to generous?

- A) cheerful
- B) helpful
- C) patient
- D) selfish

7. Rudder is to ship as steering wheel is to ____.

- A) bicycle
- B) automobile
- C) engine
- D) road

8. Scales are to weight as ruler is to ____.

- A) paper
- B) shape
- C) distance
- D) number

9. Author is to novel as sculptor is to ____.

- A) statue
- B) studio
- C) stone
- D) chisel

10. Which word does not belong with the others?

- A) orbit
- B) evaporate
- C) rotate
- D) revolve

11. Maple, birch, and oak belong to the same group. Which word belongs with them?

- A) pine
- B) fern
- C) moss
- D) rose

12. Which word does not belong with the others?

- A) paragraph
- B) fraction
- C) decimal
- D) percent

13. A workshop has three sessions: morning, midday, and afternoon. Art must occur before music. Science must occur after music. Which subject must be in the afternoon?

- A) art
- B) music
- C) It cannot be determined.
- D) science

14. Every bronze token is round. No round token is stored in Box C. Which conclusion must be true?

- A) Every token outside Box C is bronze.
- B) No bronze token is stored in Box C.
- C) All round tokens are bronze.
- D) Some bronze tokens are square.

15. Four books—J, K, L, and M—are arranged in a row. K is immediately after J. M is before J. L is after K. Which book must be first?

- A) J
- B) K
- C) L
- D) M

16. Which word is closest in meaning to frugal?

- A) economical
- B) generous
- C) careless
- D) ambitious

17. Which word is closest in meaning to diminish?

- A) discover
- B) combine
- C) reduce
- D) predict

18. Which word is closest in meaning to vigilant?

- A) generous
- B) watchful
- C) silent
- D) confused

19. Which word is most nearly opposite in meaning to rigid?

- A) narrow
- B) plain
- C) heavy
- D) flexible

20. Which word is most nearly opposite in meaning to compulsory?

- A) formal
- B) urgent
- C) difficult
- D) optional

21. Which word is most nearly opposite in meaning to temporary?

- A) permanent
- B) uncertain
- C) recent
- D) brief

22. Larva is to butterfly as tadpole is to ____.

- A) pond
- B) frog
- C) fish
- D) egg

23. Bark is to tree as shell is to ____.

- A) ocean
- B) nest
- C) sand
- D) turtle

24. Apprentice is to master as student is to ____.

- A) classroom
- B) lesson
- C) book
- D) teacher

25. Which word does not belong with the others?

- A) trumpet
- B) cello
- C) viola
- D) violin

26. Copper, iron, and aluminum belong to the same group. Which word belongs with them?

- A) clay
- B) granite
- C) glass
- D) silver

27. Which word does not belong with the others?

- A) tiny
- B) minute
- C) enormous
- D) miniature

28. A museum tour visits exactly four rooms: P, Q, R, and S. Q is visited before R. S is visited immediately before P. R is not last. Which room must be last?

- A) P
- B) R
- C) Q
- D) S

29. If the library is open, then the study room is available. The study room is not available. What must be true?

- A) The library is open.
- B) The library may or may not be open.
- C) The library is not open.
- D) The study room is open.

30. All cedar boxes are wooden. Some wooden boxes have brass hinges. Which statement is guaranteed?

- A) Every cedar box is wooden.
- B) Some cedar boxes have brass hinges.
- C) No cedar boxes have brass hinges.
- D) Every wooden box is cedar.

31. Which word is closest in meaning to meticulous?

- A) precise
- B) ordinary
- C) careless
- D) rapid

32. Which word is closest in meaning to feasible?

- A) distant
- B) possible
- C) fragile
- D) formal

33. Which word is closest in meaning to candid?

- A) secretive
- B) honest
- C) uncertain
- D) polite

34. Which word is most nearly opposite in meaning to obscure?

- A) minor
- B) unclear
- C) ancient
- D) famous

35. Which word is most nearly opposite in meaning to expand?

- A) measure
- B) contract
- C) explain
- D) arrange

36. Which word is most nearly opposite in meaning to reckless?

- A) cautious
- B) energetic
- C) curious
- D) stubborn

37. Page is to book as brick is to ____.

- A) clay
- B) roof
- C) mason
- D) wall

38. Scales are to fish as feathers are to ____.

- A) wing
- B) bird
- C) nest
- D) beak

39. Sextant is to navigator as gavel is to ____.

- A) sailor
- B) lawyer
- C) judge
- D) carpenter

40. Which word does not belong with the others?

- A) orchid
- B) tulip
- C) fern
- D) daffodil

41. Mercury, Venus, and Mars belong to the same group. Which word belongs with them?

- A) Europa
- B) Saturn
- C) Ceres
- D) Sirius

42. Which word does not belong with the others?

- A) hexagon
- B) triangle
- C) pentagon
- D) sphere

43. Four folders W, X, Y, and Z are stacked from top to bottom. W is above X. X is above Z. Y is below Z. Which folder must be on top?

- A) X
- B) Y
- C) W
- D) Z

44. Every member of the robotics team attends the safety briefing. Priya is a member of the robotics team. What must be true?

- A) Priya leads the briefing.
- B) Everyone at the briefing is on the robotics team.
- C) Priya attends the safety briefing.
- D) Priya attends every school meeting.

45. A workshop offers pottery only on Tuesday or Thursday. Elena attends pottery, but she cannot attend on Thursday. When does Elena attend?

- A) Monday
- B) Wednesday
- C) Thursday
- D) Tuesday

46. Which word is closest in meaning to resilient?

- A) flexible
- B) silent
- C) durable
- D) narrow

47. Which word is closest in meaning to concise?

- A) colorful
- B) brief
- C) complex
- D) uncertain

48. Which word is closest in meaning to verify?

- A) delay
- B) predict
- C) divide
- D) confirm

49. Which word is most nearly opposite in meaning to abundant?

- A) plentiful
- B) useful
- C) scarce
- D) visible

50. Which word is most nearly opposite in meaning to tranquil?

- A) restless
- B) distant
- C) peaceful
- D) steady

51. Which word is most nearly opposite in meaning to permit?

- A) allow
- B) refuse
- C) request
- D) observe

52. Hive is to bee as den is to ____.

- A) fox
- B) feather
- C) branch
- D) field

53. Wool is to sweater as clay is to ____.

- A) kiln
- B) soil
- C) pottery
- D) shovel

54. Hunger is to eat as fatigue is to ____.

- A) race
- B) rest
- C) listen
- D) measure

55. Which word does not belong with the others?

- A) quartz
- B) granite
- C) marble
- D) limestone

56. Copper, silver, and gold share which objective category?

- A) fabrics
- B) metals
- C) minerals used only as fuel
- D) types of wood

57. Which word belongs with canyon, valley, and gorge?

- A) ravine
- B) harbor
- C) island
- D) glacier

58. Five runners finish a race. N finishes before P but after R. T finishes after P. S finishes before R. Who must finish first?

- A) N
- B) P
- C) R
- D) S

59. If a package is marked EXPRESS, it is placed on Cart 2. This package is not on Cart 2. What must be true?

- A) It is damaged.
- B) It is on Cart 1.
- C) It is not marked EXPRESS.
- D) It arrived yesterday.

60. Exactly one of Room A and Room B is open. Room A is open. Which statement must be true?

- A) Both rooms are closed.
- B) Room B is open.
- C) Room B is closed.
- D) A third room must be open.

Section 2: Quantitative Skills

52 questions | Recommended time: 35 minutes

61. What number comes next? 6, 11, 17, 24, 32, ____

- A) 39
- B) 40
- C) 41
- D) 42

62. What number comes next? 128, 64, 32, 16, ____

- A) 12
- B) 4
- C) 6
- D) 8

63. What number comes next? 3, 7, 6, 10, 9, 13, ____

- A) 10
- B) 11
- C) 14
- D) 12

64. What number comes next? 1, 3, 9, 27, ____

- A) 81
- B) 54
- C) 72
- D) 108

65. Which value is equivalent to 35%?

- A) $\frac{3}{10}$
- B) $\frac{7}{20}$
- C) $\frac{5}{12}$
- D) $\frac{2}{5}$

66. If $\frac{4}{7}$ of a number is 28, what is the number?

- A) 49
- B) 42
- C) 45
- D) 56

67. A quantity increases from 50 to 65. What is the percent increase?

- A) 30%
- B) 25%
- C) 35%
- D) 20%

68. Quantity A: $\frac{5}{8}$ of 64

Quantity B: 40

- A) Quantity A is greater.
- B) The two quantities are equal.
- C) Quantity B is greater.
- D) The relationship cannot be determined from the information given.

69. Quantity A: x , where $3x + 2 = 20$

Quantity B: 7

- A) Quantity A is greater.
- B) The two quantities are equal.
- C) Quantity B is greater.
- D) The relationship cannot be determined from the information given.

70. Quantity A: $a - 2$

Quantity B: 5

Given that a is an integer greater than 6.

- A) The relationship cannot be determined from the information given.
- B) Quantity B is greater.
- C) The two quantities are equal.
- D) Quantity A is greater.

71. Quantity A: The area of a square with side length 7 cm

Quantity B: 48 cm^2

- A) Quantity B is greater.
- B) Quantity A is greater.
- C) The two quantities are equal.
- D) The relationship cannot be determined from the information given.

72. Quantity A: The perimeter of an equilateral triangle with side length 9 cm

Quantity B: 28 cm

- A) Quantity A is greater.
- B) Quantity B is greater.
- C) The two quantities are equal.
- D) The relationship cannot be determined from the information given.

73. Quantity A: The volume of a cube with edge length 3 cm

Quantity B: 27 cm^3

- A) The two quantities are equal.
- B) Quantity B is greater.
- C) Quantity A is greater.
- D) The relationship cannot be determined from the information given.

74. What number comes next? 2, 5, 11, 23, 47, ____

- A) 71
- B) 95
- C) 96
- D) 94

75. What number comes next? 90, 81, 73, 66, 60, ____

- A) 54
- B) 56
- C) 55
- D) 57

76. What number comes next? 4, 12, 6, 18, 9, 27, ____

- A) 12
- B) 18
- C) 54
- D) 13.5

77. A number is divided by 4 and then increased by 7 to get 16. What is the number?

- A) 9
- B) 36
- C) 28
- D) 64

78. If $4n - 9 = 35$, what is n ?

- A) 8
- B) 9
- C) 11
- D) 13

79. A ratio is 5:8. If the first quantity is 35, what is the second quantity?

- A) 40
- B) 48
- C) 64
- D) 56

80. Which expression has the greatest value?

- A) $\frac{3}{4}$
- B) 0.72
- C) 71%
- D) $\frac{7}{10}$

81. Quantity A: $2^3 + 5$

Quantity B: $3^2 + 4$

- A) Quantity A is greater.
- B) The two quantities are equal.
- C) Quantity B is greater.
- D) The relationship cannot be determined from the information given.

82. Quantity A: 40% of 75

Quantity B: $\frac{1}{3}$ of 96

- A) Quantity A is greater.
- B) The relationship cannot be determined from the information given.
- C) The two quantities are equal.
- D) Quantity B is greater.

83. Quantity A: $n + 4$

Quantity B: 12

Given that $n < 8$.

- A) Quantity A is greater.
- B) Quantity B is greater.
- C) The two quantities are equal.
- D) The relationship cannot be determined from the information given.

84. Quantity A: The diameter of a circle with radius 8 cm

Quantity B: 15 cm

- A) The relationship cannot be determined from the information given.
- B) Quantity B is greater.
- C) The two quantities are equal.
- D) Quantity A is greater.

85. Quantity A: The area of a triangle with base 10 cm and height 6 cm

Quantity B: 30 cm^2

- A) Quantity A is greater.
- B) Quantity B is greater.
- C) The relationship cannot be determined from the information given.
- D) The two quantities are equal.

86. Quantity A: The third angle of a triangle whose other angles are 35° and 75°

Quantity B: 65°

- A) Quantity B is greater.
- B) Quantity A is greater.
- C) The two quantities are equal.
- D) The relationship cannot be determined from the information given.

87. What number comes next? 1, 4, 10, 19, 31, ____

- A) 46
- B) 43
- C) 49
- D) 40

88. What number comes next? 160, 80, 40, 20, ____

- A) 18
- B) 5
- C) 15
- D) 10

89. What number comes next? 3, 8, 5, 10, 7, 12, ____

- A) 9
- B) 14
- C) 17
- D) 8

90. What number comes next? 0.2, 0.6, 1.8, 5.4, ____

- A) 16.2
- B) 10.8
- C) 12.6
- D) 18.4

91. A store reduces a \$60 item by 25%. What is the sale price?

- A) \$15
- B) \$35
- C) \$45
- D) \$50

92. Which fraction is equivalent to 0.375?

- A) $\frac{3}{8}$
- B) $\frac{5}{8}$
- C) $\frac{3}{5}$
- D) $\frac{7}{20}$

93. If 7 notebooks cost \$21 at the same unit price, how much do 5 notebooks cost?

- A) \$12
- B) \$18
- C) \$15
- D) \$20

94. Quantity A: $|-9|$

Quantity B: 3^2

- A) Quantity A is greater.
- B) Quantity B is greater.
- C) The relationship cannot be determined from the information given.
- D) The two quantities are equal.

95. Quantity A: $\frac{5}{6}$ of 42

Quantity B: 34

- A) The two quantities are equal.
- B) Quantity B is greater.
- C) Quantity A is greater.
- D) The relationship cannot be determined from the information given.

96. Quantity A: $x - 2$

Quantity B: 5

Given that $x \geq 7$.

- A) Quantity A is greater.
- B) Quantity B is greater.
- C) The relationship cannot be determined from the information given.
- D) The two quantities are equal.

97. Quantity A: The perimeter of a square with side length 7 cm

Quantity B: 30 cm

- A) Quantity B is greater.
- B) Quantity A is greater.
- C) The two quantities are equal.
- D) The relationship cannot be determined from the information given.

98. Quantity A: The volume of a rectangular prism measuring 2 cm by 3 cm by 5 cm

Quantity B: 30 cm^3

- A) Quantity A is greater.
- B) Quantity B is greater.
- C) The two quantities are equal.
- D) The relationship cannot be determined from the information given.

99. Quantity A: An angle supplementary to a 112° angle

Quantity B: 70°

- A) Quantity A is greater.
- B) The relationship cannot be determined from the information given.
- C) The two quantities are equal.
- D) Quantity B is greater.

100. What number comes next? 2, 6, 7, 21, 22, 66, ____

- A) 132
- B) 69
- C) 67
- D) 198

101. What number comes next? 81, 27, 9, 3, ____

- A) 0
- B) 1
- C) 2
- D) 1.5

102. What number comes next? 5, 7, 14, 16, 32, 34, ____

- A) 68
- B) 36
- C) 52
- D) 66

103. A number is increased by 20% and becomes 72. What was the original number?

- A) 50
- B) 64
- C) 54
- D) 60

104. Which value is equal to $\frac{3}{4} \div \frac{1}{8}$?

- A) $\frac{3}{32}$
- B) 6
- C) $\frac{3}{2}$
- D) 8

105. The ratio of red beads to blue beads is 3:5. If there are 32 beads altogether, how many are red?

- A) 8
- B) 12
- C) 20
- D) 15

106. Which number leaves a remainder of 2 when divided by 5?

- A) 48
- B) 45
- C) 42
- D) 50

107. Quantity A: $7^2 - 13$

Quantity B: 6^2

- A) The two quantities are equal.
- B) Quantity B is greater.
- C) Quantity A is greater.
- D) The relationship cannot be determined from the information given.

108. Quantity A: y

Quantity B: 12

Given that $y < 12$.

- A) Quantity A is greater.
- B) The relationship cannot be determined from the information given.
- C) The two quantities are equal.
- D) Quantity B is greater.

109. Quantity A: $2k$

Quantity B: $k + 6$

Given that k is positive.

- A) Quantity A is greater.
- B) The relationship cannot be determined from the information given.
- C) The two quantities are equal.
- D) Quantity B is greater.

110. Quantity A: The area of a circle with radius 4 cm, using 3.14 for π

Quantity B: 50 cm²

- A) The two quantities are equal.
- B) Quantity B is greater.
- C) Quantity A is greater.
- D) The relationship cannot be determined from the information given.

111. Quantity A: The hypotenuse of a right triangle with legs 5 cm and 12 cm

Quantity B: 14 cm

- A) Quantity A is greater.
- B) The two quantities are equal.
- C) Quantity B is greater.
- D) The relationship cannot be determined from the information given.

112. Quantity A: The area of a parallelogram with base 9 cm and height 4 cm

Quantity B: 36 cm²

- A) Quantity A is greater.
- B) Quantity B is greater.
- C) The relationship cannot be determined from the information given.
- D) The two quantities are equal.

Section 3: Reading Comprehension

62 questions | Recommended time: 35 minutes

Literary Passage: The Unfinished Mosaic

Tessa had volunteered to help restore the tiled sign above the entrance to the old neighborhood market. Most of the work involved cleaning intact pieces, but one corner of the mosaic was missing several tiles.

The restoration leader, Ms. Ortiz, spread boxes of replacement pieces across a table. Some matched the original colors exactly. Others were close but not identical. Tessa assumed they should use only perfect matches so the repair would disappear.

Ms. Ortiz disagreed. "If we pretend nothing ever happened to the sign, we lose part of its story," she said. She chose tiles that fit the original pattern but were slightly different in shade.

At first, Tessa thought the new corner looked wrong. The design lined up, yet the repair was visible in afternoon light. As they worked, Ms. Ortiz explained that the sign had survived storms, building renovations, and decades of changing businesses.

When the mosaic was finished, the repaired section no longer bothered Tessa. Its colors were not identical, but they belonged to the pattern without pretending to be original.

The next morning, Tessa passed the market and looked up. The repaired corner did not erase the sign's age. Instead, it showed that someone had decided the old sign was worth keeping.

113. Why does Tessa initially want exact-color replacement tiles?

- A) She dislikes the original pattern.
- B) She wants to finish faster.
- C) She wants the repair to disappear.
- D) Ms. Ortiz requires it.

114. Why does Ms. Ortiz prefer slightly different replacement tiles?

- A) They cost less.
- B) They are easier to cut.
- C) They are brighter than the originals.
- D) They preserve evidence that the sign has a history of damage and repair.

115. As used in the fifth paragraph, "belonged" most nearly means

- A) was newly purchased
- B) was owned by someone
- C) was stored nearby
- D) fit appropriately

116. How does Tessa change during the passage?

- A) She becomes less interested in restoration.
- B) She decides the old sign should be replaced.
- C) She learns that a repair can respect history without perfectly copying the original.
- D) She stops noticing differences in color.

117. Which statement best expresses a central idea?

- A) Old objects should always remain damaged.
- B) Preserving something can include acknowledging the changes it has undergone.
- C) Repairs are only successful when invisible.
- D) New materials are always better than old ones.

Informational Passage: Why Some Bridges Have Expansion Joints

A bridge may appear solid and motionless, but its materials expand and contract as temperatures change. Steel and concrete become slightly larger when warmed and slightly smaller when cooled.

If a long bridge had no room for this movement, repeated temperature changes could place stress on its structure. Engineers therefore include expansion joints at selected locations. These joints create small gaps or flexible connections that allow parts of the bridge to move without pushing dangerously against one another. The amount of movement depends on several factors, including the bridge's length, the materials used, and the range of temperatures expected in the region. A short bridge in a mild climate may require less movement space than a much longer bridge exposed to very hot summers and cold winters.

Expansion joints must also handle traffic. Cars and trucks pass over them repeatedly, so joints are designed to support loads while still allowing controlled motion. They require inspection because dirt, ice, or worn parts can reduce how well they function.

The joints are therefore not signs of weakness. They are deliberate features that help a rigid-looking structure respond safely to natural changes in temperature.

118. What causes bridge materials to expand and contract?

- A) Traffic speed
- B) Wind direction
- C) Paint color
- D) Temperature changes

119. As used in the second paragraph, “stress” most nearly means

- A) structural force or strain
- B) worry
- C) noise
- D) weight alone

120. Why might a very long bridge require larger movement allowances?

- A) More material can undergo temperature-related expansion and contraction.
- B) It is always built from weaker materials.
- C) Long bridges receive more sunlight at night.
- D) It carries fewer vehicles.

121. How is the passage mainly organized?

- A) It compares bridges from different centuries.
- B) It lists bridge materials from strongest to weakest.
- C) It tells a story about one bridge engineer.
- D) It explains a physical problem, describes an engineering solution, then discusses design and maintenance factors.

122. Why must expansion joints be inspected?

- A) They change bridge color.
- B) They may become blocked or worn.
- C) They measure traffic speed.
- D) They must be removed in winter.

Reflective Passage: What Makes a Useful Archive?

An archive is often imagined as a place filled with rare documents: signed letters, old photographs, official records, and carefully preserved books. Those materials can be valuable, but an archive's usefulness depends on more than the age or rarity of what it contains.

Researchers need context. A single photograph may show a crowded street, yet without a date, location, or

explanation, its meaning can be difficult to determine. A simple receipt or handwritten schedule may seem ordinary, but if it identifies a business, event, or daily routine, it can help researchers understand a larger pattern. Organization also matters. Materials that cannot be located are difficult to study. Archivists therefore create descriptions, labels, and finding aids that connect individual items to people, places, dates, and subjects. Preservation is another part of the work. Documents must be protected from moisture, light, pests, and rough handling. Increasingly, archives also preserve digital files, which create different challenges because formats and storage systems can become outdated.

A useful archive, then, is not merely a room full of old objects. It is a collection whose materials are preserved, organized, and described well enough for people to ask meaningful questions about the past.

123. What is the main idea of the passage?

- A) Archives are useful when materials are preserved, organized, and given enough context to support research.
- B) Only rare documents belong in archives.
- C) Digital files cannot be archived.
- D) Photographs are the most useful archival sources.

124. As used in the third paragraph, “finding aids” most nearly refers to

- A) devices for repairing documents
- B) machines that copy photographs
- C) shelves used for rare books
- D) tools that help researchers locate and understand materials

125. Why might an ordinary receipt be valuable to a researcher?

- A) It is always older than a photograph.
- B) It is easier to preserve than other documents.
- C) It can provide evidence about businesses, events, or daily life.
- D) It is usually signed by a famous person.

126. Which best describes the author's tone?

- A) dismissive
- B) analytical and explanatory
- C) humorous
- D) angry

127. Which preservation challenge is specifically mentioned for digital files?

- A) They always require printing.
- B) They may become too heavy.
- C) They cannot be labeled.
- D) Formats and storage systems can become outdated.

128. Which conclusion is best supported by the passage?

- A) Good organization can increase the research value of ordinary materials.
- B) Archives should avoid digital materials.
- C) Researchers need only one kind of evidence.
- D) An item becomes useful only when it is rare.

Literary Passage: The Last Rehearsal

Mara had practiced the opening of her piano duet so many times that she could hear every measure before her fingers touched the keys. Yet during the final rehearsal, her partner, Jonah, stopped halfway through the second page.

"I keep rushing this part," he said, pointing to a row of quick notes.

Mara glanced at the clock. Their recital was the next evening, and she wanted one flawless run before leaving. She suggested starting again from the beginning.

Their teacher, Mr. Bell, shook his head. Instead, he asked them to play only the troublesome four measures, first at half speed. Mara felt impatient. The opening was already polished, and repeating a tiny section seemed less satisfying than completing the whole piece.

After several slow attempts, however, Jonah stopped rushing. More surprisingly, Mara noticed that she had been entering one beat too early after his phrase. The mistake had been small enough to hide whenever they played quickly.

They practiced the four measures again, then connected them to the music before and after. On the final run, neither student tried to race toward the ending. The duet felt steadier than it ever had.

As Mara packed her music, she realized that a complete performance had not been what they needed most. Their best rehearsal had come from spending time on the smallest unfinished part.

129. What problem does Jonah identify during rehearsal?

- A) He forgets the opening.
- B) He plays one section too quickly.
- C) He cannot read the final page.
- D) He enters one beat late.

130. Why is Mara initially dissatisfied with practicing only four measures?

- A) She believes a complete run would feel more useful and satisfying.
- B) She wants Jonah to practice alone.
- C) She cannot play slowly.
- D) She dislikes the piece.

131. What role does the four-measure practice play in the plot?

- A) It introduces a new character.
- B) It causes Mara to cancel the recital.
- C) It proves the opening is incorrect.
- D) It reveals a hidden problem and changes how the students rehearse.

132. Which idea is best supported by the passage?

- A) Careful attention to a weak section can improve the larger performance.
- B) Finishing an entire task is always more useful than focusing on details.
- C) Musicians should never practice at full speed.
- D) Partners usually make the same mistakes.

133. As used in the fifth paragraph, "polished" most nearly means

- A) shiny
- B) rewritten
- C) expensive
- D) well practiced

Scientific Passage: How Desert Beetles Collect Water

In extremely dry environments, survival may depend on obtaining water from sources that are easy to overlook. Some desert beetles can collect tiny droplets from fog that moves across coastal dunes.

A beetle may climb to an exposed position when fog arrives and angle its body so moisture can gather on its outer surface. Microscopic features on that surface help droplets form and combine. As droplets grow, gravity

can pull them toward the beetle's mouth.

Researchers study this process because it demonstrates how surface structure can influence the movement of water. A material does not need to absorb water like a sponge in order to collect it. Instead, differences in texture and surface properties can encourage droplets to form in particular places and then move in a useful direction. Engineers have experimented with materials inspired by similar principles. Possible applications include surfaces designed to capture moisture from air in dry regions. However, copying a biological system is not as simple as reproducing its appearance. A useful engineered surface must also work under realistic conditions, be durable, and collect enough water to justify its cost.

The beetle's strategy is therefore valuable not because it offers a ready-made machine, but because it reveals a physical principle that designers can investigate and adapt.

134. What helps water droplets move toward the beetle's mouth?

- A) Sunlight
- B) Gravity
- C) Sand
- D) Wind alone

135. What is the main purpose of the passage?

- A) To argue that all deserts contain heavy fog
- B) To compare several desert insects
- C) To explain a beetle's water-collection strategy and why it interests engineers
- D) To describe how beetles store water underground

136. Why does the author say copying the beetle is not simply a matter of copying its appearance?

- A) Engineered designs must also function effectively and practically.
- B) Beetles change color in fog.
- C) The beetle's surface cannot be studied.
- D) Artificial materials always absorb too much water.

137. How does the passage develop its explanation?

- A) It lists beetle species alphabetically.
- B) It describes a natural process, explains the principle behind it, and then discusses possible engineering applications.
- C) It presents an argument against studying insects.
- D) It tells the history of one engineering company.

138. According to the passage, what can microscopic surface features help droplets do?

- A) Evaporate instantly
- B) Form and combine
- C) Turn into ice
- D) Enter the sand

Informational Passage: The Quiet Work of Seed Banks

A seed bank stores seeds so that plant varieties can be preserved for future study and use. Collections may include seeds from crops, wild relatives of crops, and plants adapted to particular climates or soils.

Storage begins long before a container is placed on a shelf. Staff members document where seeds came from and identify the plant variety as accurately as possible. Many seeds are dried to reduce moisture and then kept at low temperatures, conditions that can slow biological processes and extend storage life.

Stored seeds do not remain useful forever without attention. Workers periodically test samples to see how many seeds still germinate. If germination rates fall too low, staff may grow plants from the remaining viable seeds and

collect a fresh generation for storage. This process must be managed carefully so that the renewed sample continues to represent the original variety.

Seed banks are sometimes described as insurance against crop loss, but their role is broader. Researchers can use stored material to study traits such as disease resistance, drought tolerance, or adaptation to local conditions. Plant breeders may also use preserved varieties when developing new crops.

A seed bank is therefore both a storage facility and an active scientific resource. Its value depends not merely on keeping seeds cold, but on maintaining accurate records, monitoring viability, and preserving useful genetic diversity over time.

139. Which statement best expresses the central idea?

- A) Seed banks are mainly museums for rare containers.
- B) All stored seeds remain viable indefinitely.
- C) Seed banks require active management to preserve useful plant diversity for future research and use.
- D) Only crop seeds can be stored in seed banks.

140. Why are many seeds dried before cold storage?

- A) To make them easier to identify
- B) To reduce moisture and help extend storage life
- C) To change their genetic traits
- D) To make them germinate immediately

141. Why must staff be careful when producing a fresh generation of stored seeds?

- A) All new plants must grow to the same height.
- B) Seeds can only be renewed once.
- C) The renewed sample should continue to represent the original variety.
- D) The new seeds must be larger than the originals.

142. Why does the author use the phrase “insurance against crop loss”?

- A) To present a familiar comparison for one protective function of seed banks
- B) To suggest farmers pay monthly fees
- C) To claim seed banks sell financial policies
- D) To criticize seed storage

143. Which best summarizes the final two paragraphs?

- A) Seed banks should stop storing wild plants.
- B) Seed banks support research and breeding, and their usefulness depends on active preservation practices.
- C) Cold storage is the only task performed by seed banks.
- D) Plant breeders rarely use preserved material.

144. What do workers periodically test in stored seed samples?

- A) Their germination rate
- B) Their market price
- C) Their color
- D) Their container weight

Literary Passage: The Missing Label

Avery had volunteered to organize the props for the school play because the job seemed straightforward. Each object had a shelf, each shelf had a label, and the cast had been told to return everything after rehearsal.

On Wednesday, however, the brass-colored lantern used in the second scene disappeared. Avery searched the prop room twice and then began questioning cast members. No one remembered moving it.

With opening night only two days away, Avery considered asking the art club to make a replacement. Before doing so, she returned to the stage and watched the crew practice a scene change. A student carried a wooden crate offstage, and Avery noticed a strip of gold through a gap in the boards.

The lantern was inside.

The crew chief explained that the crate had become a temporary holding place for objects needed during quick scene changes. No one had told Avery because everyone assumed the arrangement was obvious.

Avery brought the lantern back after rehearsal, but she did not simply restore the old system. She made a new label for the crate and added it to her prop checklist. The next day, she also asked the crew chief to tell her whenever the location of a prop changed.

The lantern had not really been lost. The information about where it belonged had been.

145. Where does Avery eventually find the lantern?

- A) In the art room
- B) On its labeled shelf
- C) Inside a wooden crate
- D) Under the stage

146. Why had no one told Avery about the crate?

- A) The crew wanted to hide the lantern.
- B) People assumed the temporary arrangement was obvious.
- C) The cast believed Avery had made the change.
- D) The crate had no connection to the play.

147. What does the missing lantern mainly reveal in the story?

- A) A disagreement about the play's design
- B) A shortage of stage equipment
- C) A communication problem between people using different systems
- D) A reason to cancel rehearsal

148. Which idea is best supported by the passage?

- A) Temporary solutions should never be used.
- B) Clear communication can be as important as careful organization.
- C) Checklists cause confusion during performances.
- D) Stage crews work best without labels.

149. As used in the final paragraph, "belonged" most nearly means

- A) was owned by someone
- B) was permanently attached
- C) was valuable
- D) was supposed to be located

Literary Passage: A Different Route

Mr. Chen's photography club usually walked the same path through the neighborhood park. The route passed a pond, a stone bridge, and a row of old sycamores, all dependable subjects for beginning photographers.

One Saturday, construction closed the bridge. Several students groaned when Mr. Chen led them along a narrow side trail instead. The path seemed ordinary at first, bordered by shrubs and a plain wooden fence.

Then Lila noticed sunlight passing through a tear in one leaf, casting a bright oval on the fence. She crouched to photograph it. Farther ahead, another student found repeating shadows made by bicycle spokes at a repair station. Soon the group was stopping every few steps.

Mr. Chen did not point out subjects. Instead, he asked questions: Where does your eye go first? What changes if you move two steps to the left? Which detail would someone else overlook?

By the time the group reached the far side of the park, no one was complaining about the detour. Lila had taken fewer photographs than usual, but she had spent more time deciding what each picture should include.

At the next meeting, Mr. Chen displayed photographs from both the familiar route and the detour. The strongest images, he observed, were not tied to the most impressive scenery. They came from photographers who had learned to notice deliberately.

150. Why does the club take a different path?

- A) The pond is flooded.
- B) Mr. Chen wants to avoid the sycamores.
- C) Construction has closed the bridge.
- D) The students request a longer walk.

151. What can be inferred about Mr. Chen's questions?

- A) They test whether students memorized camera instructions.
- B) They are meant to speed up the walk.
- C) They identify the only subjects worth photographing.
- D) They encourage students to make intentional visual choices.

152. What lesson does the club's detour illustrate?

- A) Photographers should always take as many pictures as possible.
- B) Good photographs require famous landmarks.
- C) Careful observation can make ordinary surroundings visually interesting.
- D) Unexpected routes are usually shorter.

153. How does the passage mainly develop?

- A) It compares two camera brands.
- B) It moves from disappointment about a detour to discovery and reflection.
- C) It lists park features from largest to smallest.
- D) It explains construction procedures.

154. Why does the author mention Lila taking fewer photographs than usual?

- A) To show her camera is malfunctioning
- B) To prove the detour has fewer subjects
- C) To suggest she dislikes photography
- D) To contrast quantity with more thoughtful decision-making

Informational Passage: Why Some Bridges Use Expansion Joints

A bridge may appear solid and motionless, yet its materials change slightly as temperatures rise and fall. Many materials expand when warmed and contract when cooled. Over a long structure, even small changes in length can matter.

Engineers account for this movement in several ways. One common method is to include expansion joints, which are designed spaces or flexible connections between sections of a bridge. These joints allow parts of the structure to move a limited amount without forcing the entire bridge to resist that movement.

The amount of expected expansion depends on factors such as the bridge's materials, its length, and the range of temperatures at its location. A short bridge in a mild climate may face different demands from a long bridge exposed to large seasonal temperature changes.

Expansion joints also require maintenance. Dirt and debris can collect in them, and repeated traffic can wear

their components. If a joint no longer moves as intended, stress may be transferred to other parts of the structure.

For this reason, expansion joints are not simply gaps left during construction. They are planned features that help engineers manage predictable movement. Their design illustrates a broader engineering principle: structures can sometimes become more durable when they are allowed to move in controlled ways rather than being made completely rigid.

155. What is one purpose of an expansion joint?

- A) To allow limited movement between bridge sections
- B) To increase traffic speed
- C) To keep bridge materials permanently warm
- D) To prevent all bridge movement

156. What is the central idea of the passage?

- A) Rigid bridges never require maintenance.
- B) All bridges expand by exactly the same amount.
- C) Bridge movement is caused mainly by traffic.
- D) Expansion joints help bridges manage temperature-related movement and must be designed and maintained carefully.

157. Why might two bridges require different expansion-joint designs?

- A) Only one bridge may carry vehicles.
- B) Expansion joints are decorative features.
- C) Engineers choose joint designs randomly.
- D) Their expected movement may differ because of material, length, and climate.

158. As used in the fourth paragraph, “transferred” most nearly means

- A) measured
- B) moved from one place to another
- C) eliminated
- D) hidden

159. Which statement best summarizes the final paragraph?

- A) Expansion joints are accidental construction spaces.
- B) Controlled flexibility can help a structure handle predictable movement.
- C) Maintenance can eliminate temperature changes.
- D) Bridge materials should never change length.

Literary Passage: The Quiet Rehearsal

Nadia expected the auditorium to be noisy on the afternoon before the debate showcase. Instead, when she arrived, she found her partner Eli sitting alone in the front row with a stack of index cards.

“We already know our arguments,” Nadia said. “Shouldn’t we practice speaking?”

Eli handed her a card. On it he had written only one sentence: What would make someone disagree with us? Nadia frowned. Their presentation supported adding more shaded seating to the school courtyard. They had gathered temperature measurements, surveyed students, and prepared photographs. She had assumed the evidence spoke for itself.

For the next half hour, they did not rehearse the presentation. They listed objections instead. Some students might prefer open space for games. Additional structures could cost more than expected. Maintenance workers might need clear access to certain areas.

Then Nadia and Eli revised their proposal. They suggested movable shade structures rather than permanent

ones and added a sentence acknowledging the cost concern. When they finally practiced aloud, Nadia noticed that the presentation felt stronger, not weaker.

At the showcase the next day, a judge asked about preserving space for activities. Nadia was ready. She answered without rushing and explained the movable design.

Afterward, Eli gathered the index cards. "Good thing we practiced the questions we didn't want," he said. Nadia smiled. The quietest rehearsal had prepared them for the loudest part.

160. What concern do Nadia and Eli add to their revised proposal?

- A) The courtyard closes too early.
- B) Students dislike outdoor seating.
- C) The debate showcase may be canceled.
- D) Shade structures could cost more than expected.

161. Why does Nadia initially say the evidence should speak for itself?

- A) She has forgotten the presentation.
- B) She wants Eli to gather different evidence.
- C) She believes their research already makes the proposal convincing.
- D) She doubts the temperature measurements.

162. What is the main function of Eli's index-card question in the story?

- A) It reveals that Eli wants to cancel the showcase.
- B) It changes the proposal's main topic.
- C) It proves the students collected inaccurate data.
- D) It introduces the need to consider opposing viewpoints.

163. As used in the passage, "acknowledging" most nearly means

- A) recognizing
- B) ignoring
- C) solving completely
- D) measuring

164. How is the passage mainly organized?

- A) A problem in preparation leads to a revised strategy that succeeds during the showcase.
- B) Two proposals are compared without a conclusion.
- C) Several unrelated debate techniques are listed.
- D) The events are presented in reverse chronological order.

Scientific Passage: How Desert Plants Limit Water Loss

Plants living in dry environments face a difficult problem: they need sunlight and carbon dioxide for photosynthesis, yet exchanging gases with the air can also allow water vapor to escape. Desert plants have evolved several features that help manage this tradeoff.

Many have a waxy outer coating that slows evaporation from exposed surfaces. Some also have small or narrow leaves, reducing the surface area from which water can be lost. In certain species, leaves are replaced by spines while green stems perform much of the photosynthesis.

Another adaptation involves stomata, tiny openings that regulate gas exchange. Most plants open their stomata mainly during the day. Some desert plants, however, open them primarily at night, when temperatures are lower and the air may be less drying. They store carbon-containing compounds produced from the nighttime gas exchange and use them during daylight for photosynthesis.

These adaptations do not eliminate the need for water. Roots must still absorb moisture, and prolonged drought can stress even highly adapted species. Instead, the features reduce unnecessary water loss and help plants

make use of limited supplies.

A desert plant's survival therefore depends not on a single unusual trait but on several coordinated features.

Shape, surface coverings, gas-exchange timing, and root function can all contribute to conserving a resource that is difficult to replace.

165. When do some desert plants primarily open their stomata?

- A) At midday
- B) During rainfall only
- C) At night
- D) Only in winter

166. What is the main idea of the passage?

- A) All desert plants survive without roots.
- B) Desert plants use several coordinated adaptations to reduce water loss while carrying out necessary functions.
- C) Spines are the only important desert adaptation.
- D) Photosynthesis can occur only at night in deserts.

167. Why can opening stomata at night help conserve water?

- A) Nighttime conditions are generally cooler and may reduce evaporation.
- B) Plants absorb sunlight more efficiently at night.
- C) Roots stop functioning during daylight.
- D) Night air contains no water vapor.

168. As used in the first paragraph, “tradeoff” most nearly means

- A) a balance involving competing benefits and costs
- B) a sudden weather change
- C) an exchange of one plant for another
- D) a type of root growth

169. Why does the author discuss several different adaptations?

- A) To prove every desert plant has identical structures
- B) To show that water conservation can depend on multiple features working together
- C) To explain how deserts receive rainfall
- D) To rank desert plants from strongest to weakest

Humanities Passage: A Mapmaker's Field Notes

Before satellite images and digital mapping tools, mapmakers often depended on measurements collected directly in the field. A surveyor might record distances, directions, elevations, landmarks, road conditions, and the positions of rivers or buildings. These observations were then compared and converted into maps.

Field notebooks were essential to this work. A page might contain neat columns of measurements beside hurried sketches, abbreviations, weather notes, or reminders to check a doubtful reading. Such notebooks were working documents rather than polished publications.

Their imperfections can make them valuable to historians. A finished map shows the result of a mapping project, but field notes may reveal how that result was produced. Crossed-out numbers can show that a measurement was reconsidered. Repeated visits to one location may suggest that the terrain was difficult to survey. Notes about a washed-out road or a newly built bridge can also preserve details about a landscape at a particular moment.

Historians must still interpret these records carefully. A surveyor could make an error, use an unfamiliar abbreviation, or omit information that seemed obvious at the time. One notebook should not automatically be treated as complete evidence.

Even so, field notes provide something a finished map often cannot: a record of decisions, uncertainty, and revision. They show mapping not simply as the creation of a product, but as a process of observing and checking the world.

170. What might a surveyor record in a field notebook?

- A) Only the names of map buyers
- B) Only the final printed map
- C) Measurements, sketches, and reminders
- D) Instructions for operating satellites

171. Why might repeated visits to one location interest a historian?

- A) They may indicate that the area was difficult to survey.
- B) They prove the surveyor lived there.
- C) They show the finished map was never created.
- D) They mean the location changed countries.

172. As used in the fourth paragraph, “omit” most nearly means

- A) underline
- B) translate
- C) leave out
- D) correct

173. Why are field notes useful to historians according to the passage?

- A) They are always more accurate than finished maps.
- B) They contain complete records of every landscape.
- C) They can reveal the decisions and revisions behind a finished map.
- D) They eliminate the need to interpret historical evidence.

174. What effect does the contrast between field notes and finished maps create?

- A) It highlights the difference between a working process and its polished result.
- B) It suggests maps should contain crossed-out numbers.
- C) It shows that surveyors preferred writing to mapping.
- D) It proves field notes were intended for publication.

Section 4: Mathematics

64 questions | Recommended time: 45 minutes

175. Evaluate: $24 \div 6 + 3 \times 5$

- A) 20
- B) 19
- C) 23
- D) 35

176. Which fraction is equivalent to 0.45?

- A) $\frac{9}{20}$
- B) $\frac{4}{9}$
- C) $\frac{5}{11}$
- D) $\frac{11}{20}$

177. What is $|-9| + |-4|$?

- A) 5
- B) 13
- C) -13
- D) 36

178. Which number is composite?

- A) 31
- B) 37
- C) 49
- D) 41

179. If $5x - 7 = 28$, what is x ?

- A) 5
- B) 7
- C) 6
- D) 8

180. Which expression represents “three more than four times a number n ”?

- A) $3n + 4$
- B) $4(n + 3)$
- C) $7n$
- D) $4n + 3$

181. If $y = 2x + 5$ and $x = 4$, what is y ?

- A) 13
- B) 9
- C) 11
- D) 18

182. A service charges a \$12 fixed fee plus \$6 per hour. Which expression gives the total cost C for h hours?

- A) $C = 18h$
- B) $C = 12 + 6h$
- C) $C = 6 + 12h$
- D) $C = 12h + 6$

183. A triangle has angles measuring 48° , 67° , and x° . What is x ?

- A) 55°
- B) 75°
- C) 65°
- D) 85°

184. A rectangle has length 14 cm and width 5 cm. What is its area?

- A) 19 cm^2
- B) 70 cm^2
- C) 38 cm^2
- D) 140 cm^2

185. A circle has a radius of 6 cm. Using 3.14 for π , what is its circumference?

- A) 113.04 cm
- B) 18.84 cm
- C) 75.36 cm
- D) 37.68 cm

186. A road is 3.2 kilometers long. How many meters is this?

- A) 32
- B) 3,200
- C) 32,000
- D) 320

187. A recipe requires 24 ounces of juice. How many cups is that if 1 cup = 8 ounces?

- A) 2 cups
- B) 6 cups
- C) 4 cups
- D) 3 cups

188. The data set is 8, 11, 11, 13, 17. What is the mode?

- A) 11
- B) 13
- C) 8
- D) 17

189. A bag contains 3 green, 4 blue, and 5 red marbles. What is the probability of selecting a blue marble?

- A) $\frac{1}{4}$
- B) $\frac{1}{3}$
- C) $\frac{5}{12}$
- D) $\frac{2}{3}$

190. The mean of five numbers is 14. Four of the numbers are 10, 12, 15, and 18. What is the fifth number?

- A) 13
- B) 15
- C) 16
- D) 14

191. What is the greatest common factor of 54 and 90?

- A) 18
- B) 9
- C) 27
- D) 36

192. Which value is greatest?

- A) -2.4
- B) -2.14
- C) -2.41
- D) -2.04

193. Estimate 19.8×4.1 by rounding each factor to the nearest whole number.

- A) 60
- B) 100
- C) 120
- D) 80

194. Simplify: $3(2x + 5) - 4x$

- A) $10x + 15$
- B) $6x + 1$
- C) $2x + 15$
- D) $2x + 5$

195. Which value of x satisfies $2(x - 3) = x + 5$?

- A) 8
- B) 10
- C) 9
- D) 11

196. A cyclist travels 42 miles in 3 hours at a constant rate. At that rate, how far will the cyclist travel in 5 hours?

- A) 56 miles
- B) 60 miles
- C) 84 miles
- D) 70 miles

197. Which inequality represents all numbers x that are at least 6?

- A) $x < 6$
- B) $x \geq 6$
- C) $x > 6$
- D) $x \leq 6$

198. A right triangle has legs of 9 cm and 12 cm. What is the length of the hypotenuse?

- A) 15 cm
- B) 13 cm
- C) 18 cm
- D) 21 cm

199. The exterior angle of a regular hexagon measures

- A) 45°
- B) 90°
- C) 60°
- D) 120°

200. Two similar rectangles have corresponding lengths 6 cm and 15 cm. If the smaller rectangle's width is 4 cm, what is the larger rectangle's width?

- A) 10 cm
- B) 9 cm
- C) 8 cm
- D) 12 cm

201. A movie begins at 6:47 p.m. and ends at 8:35 p.m. How long does it last?

- A) 1 hour 38 minutes
- B) 2 hours 12 minutes
- C) 1 hour 48 minutes
- D) 2 hours 48 minutes

202. A package weighs 5 pounds 8 ounces. How many ounces does it weigh in all?

- A) 80 ounces
- B) 72 ounces
- C) 88 ounces
- D) 96 ounces

203. On a map, 1 inch represents 18 miles. Two towns are 3.5 inches apart on the map. What is their actual distance?

- A) 54 miles
- B) 81 miles
- C) 72 miles
- D) 63 miles

204. The numbers of pages read by five students are 18, 22, 24, 24, and 37. What is the median?

- A) 22
- B) 25
- C) 24
- D) 37

205. A spinner has 8 equal sections: 3 yellow, 2 green, and 3 purple. What is the probability of landing on a section that is not green?

- A) $\frac{3}{4}$
- B) $\frac{3}{8}$
- C) $\frac{5}{8}$
- D) $\frac{1}{4}$

206. A team scored 12, 15, 9, and 18 points in four games. How many points must it score in a fifth game to have a mean of 14 points?

- A) 20
- B) 14
- C) 18
- D) 16

207. Which number is prime?

- A) 39
- B) 51
- C) 57
- D) 53

208. What is the value of $6 + 3 \times (8 - 5)^2$?

- A) 33
- B) 45
- C) 81
- D) 87

209. Which expression is equivalent to $2\frac{3}{5}$?

- A) 2.35
- B) 2.6
- C) 3.5
- D) 2.8

210. What is 35% of 240?

- A) 96
- B) 72
- C) 84
- D) 108

211. If $f(x) = 3x - 2$, what is $f(7)$?

- A) 17
- B) 21
- C) 19
- D) 23

212. Solve: $4x + 7 = 3x + 18$

- A) 9
- B) 11
- C) 10
- D) 25

213. A taxi charges a \$4 starting fee plus \$2 for each mile traveled. Which expression gives the total cost for m miles?

- A) $4m + 2$
- B) $6m$
- C) $2m + 4$
- D) $2(m + 4)$

214. The rule $y = 5x + 1$ relates x and y . If $y = 36$, what is x ?

- A) 5
- B) 7
- C) 6
- D) 8

215. A rectangle has area 96 square units and width 8 units. What is its length?

- A) 12 units
- B) 14 units
- C) 10 units
- D) 16 units

216. Two angles of a triangle measure 48° and 67° . What is the third angle?

- A) 55°
- B) 75°
- C) 60°
- D) 65°

217. A circle has diameter 18 inches. What is its radius?

- A) 9 inches
- B) 8 inches
- C) 36 inches
- D) 6 inches

218. A container holds 2.75 liters. How many milliliters is that?

- A) 2,075
- B) 275
- C) 2,750
- D) 27,500

219. A rectangular garden is drawn at a scale of 1 cm = 4 m. If its length on the drawing is 6.5 cm, what is the actual length?

- A) 26 m
- B) 24 m
- C) 10.5 m
- D) 32 m

220. A data set is 6, 8, 8, 11, 13, 14. What is the range?

- A) 6
- B) 9
- C) 8
- D) 14

221. A bag contains 4 red, 3 blue, and 5 white marbles. What is the probability of choosing a blue marble and then, without replacement, choosing another blue marble?

- A) $\frac{1}{16}$
- B) $\frac{1}{12}$
- C) $\frac{1}{11}$
- D) $\frac{1}{22}$

222. The mean of five numbers is 18. Four of the numbers are 12, 17, 20, and 21. What is the fifth number?

- A) 18
- B) 20
- C) 19
- D) 22

223. Which number has exactly three positive factors?

- A) 18
- B) 27
- C) 30
- D) 25

224. Which value is closest to $\sqrt{70}$?

- A) 7
- B) 9
- C) 8
- D) 10

225. A jar is $\frac{3}{5}$ full. What percent of the jar is empty?

- A) 20%
- B) 40%
- C) 60%
- D) 80%

226. Simplify: $5a - 2(3a - 4)$

- A) $a + 8$
- B) $11a - 8$
- C) $-a - 8$
- D) $-a + 8$

227. Solve: $3(x + 4) = 2x + 19$

- A) 7
- B) 11
- C) 5
- D) 15

228. Which inequality is equivalent to $2x - 5 < 9$?

- A) $x < 7$
- B) $x > 2$
- C) $x > 7$
- D) $x < 2$

229. A tank contains 18 gallons and fills at 4 gallons per minute. Which equation gives the number of gallons g after t minutes?

- A) $g = 4t - 18$
- B) $g = 18t + 4$
- C) $g = 18 + 4t$
- D) $g = 22t$

230. A rectangular prism has length 8 cm, width 3 cm, and height 4 cm. What is its surface area?

- A) 136 cm^2
- B) 104 cm^2
- C) 112 cm^2
- D) 192 cm^2

231. A square has a diagonal relationship forming a right triangle with legs 6 units and 6 units. Which expression gives the diagonal length?

- A) $6\sqrt{2}$
- B) 12
- C) 6
- D) 18

232. The sum of the interior angles of a pentagon is

- A) 360°
- B) 450°
- C) 540°
- D) 720°

233. A runner completes 1.5 miles. How many yards is that if 1 mile = 1,760 yards?

- A) 880 yards
- B) 2,640 yards
- C) 1,760 yards
- D) 3,520 yards

234. A container holds 3 quarts of liquid. How many cups is that if 1 quart = 4 cups?

- A) 7 cups
- B) 12 cups
- C) 16 cups
- D) 24 cups

235. A machine produces 45 parts in 15 minutes at a constant rate. How many parts does it produce in 1 hour?

- A) 90
- B) 135
- C) 180
- D) 225

236. The values 4, 7, 9, 12, and 18 have what median?

- A) 7
- B) 10
- C) 12
- D) 9

237. A box contains 2 black, 5 white, and 3 gray tiles. One tile is chosen at random. What is the probability it is black or gray?

- A) $\frac{4}{5}$
- B) $\frac{1}{5}$
- C) $\frac{3}{5}$
- D) $\frac{1}{2}$

238. A class recorded quiz scores of 70, 80, 80, 90, and 100. If the 100 is removed, how does the mean change?

- A) It decreases by 4 points.
- B) It decreases by 5 points.
- C) It stays the same.
- D) It increases by 4 points.

Section 5: Language

60 questions | Recommended time: 30 minutes

239. Which sentence is punctuated correctly?

- A) Before the concert began the audience grew quiet.
- B) Before, the concert began the audience grew quiet.
- C) Before the concert began, the audience grew quiet.
- D) Before the concert, began the audience grew quiet.

240. Which sentence correctly uses a colon?

- A) The kit contained three: items rope, tape, and gloves.
- B) The kit contained: rope, tape, and gloves.
- C) The kit: contained three items, rope, tape, and gloves.
- D) The kit contained three items: rope, tape, and gloves.

241. Which sentence correctly uses a semicolon?

- A) The lights; dimmed, and the performance began.
- B) The lights dimmed; and the performance began.
- C) The lights dimmed; the performance began.
- D) The lights dimmed the performance; began.

242. Which sentence uses capitalization correctly?

- A) We studied french history in March.
- B) We studied French History in march.
- C) We studied French history in March.
- D) We studied french History in March.

243. Which sentence uses capitalization correctly?

- A) The Class visited Yellowstone national park on Monday.
- B) The class visited yellowstone National Park on monday.
- C) The class visited Yellowstone National park on Monday.
- D) The class visited Yellowstone National Park on Monday.

244. Which word is spelled correctly?

- A) neccessary
- B) necessary
- C) nessessary
- D) nesessary

245. Which sentence contains a misspelled word?

- A) The speaker gave a relevent example.
- B) The hikers carried sufficient water.
- C) The machine required immediate repair.
- D) The manager reviewed the schedule.

246. Which sentence has correct subject-verb agreement?

- A) The stack of boxes are near the door.
- B) A group of musicians perform every evening.
- C) The list of names were posted outside.
- D) Each of the players wears a numbered jersey.

247. Which sentence uses the correct verb form?

- A) By noon, the workers had finished the repair.
- B) By noon, the workers have finished yesterday.
- C) By noon, the workers had finish the repair.
- D) By noon, the workers finishing the repair.

248. Which sentence places the modifier correctly?

- A) Diego noticed, walking across the plaza, the fountain.
- B) Walking across the plaza, the fountain caught Diego's attention.
- C) Walking across the plaza, Diego noticed the fountain.
- D) The fountain, walking across the plaza, caught Diego's attention.

249. Which sentence uses the correct word?

- A) The storm may effect the schedule.
- B) The storm may effected the schedule.
- C) The storm may affect the schedule.
- D) The storm may affecting the schedule.

250. Which sentence is most concise?

- A) The final outcome was completely unanimous among everyone.
- B) All of everyone agreed in a unanimous way.
- C) The agreement was unanimous among everyone together.
- D) Everyone agreed unanimously.

Editing Passage: A Borrow-and-Return Station

(1) Students at Harbor Middle School often needed small supplies such as rulers, calculators, and headphones for short periods. (2) Teachers noticed that students sometimes bought inexpensive items they would use only once or twice. (3) The student council proposed a borrow-and-return station near the main office. (4) Each item received a number, and students signed a simple checkout sheet before borrowing it. (5) During the first week, several items were returned late. (6) The council added a reminder board showing due times and asked homeroom teachers to mention the system. (7) By the end of the month, most items were being returned on time. (8) The station did not eliminate the need to own supplies, but it reduced unnecessary purchases for occasional needs.

251. Which sentence would be the best topic sentence to place before sentence 1?

- A) A simple lending system helped students share occasionally needed supplies.
- B) Some students enjoy buying new materials.
- C) School offices often contain many desks.
- D) Calculators are useful in mathematics.

252. Which sentence could best be added after sentence 6?

- A) Many students also use backpacks.
- B) The board was made from blue paper.
- C) Homeroom periods begin at different times.
- D) These changes made expectations clearer for borrowers.

253. Which revision best combines sentences 7 and 8?

- A) By the end of the month, items were on time, and supplies still existed.
- B) By the end of the month, most items were returned on time, and the station reduced unnecessary purchases even though students still needed to own some supplies.
- C) The station eliminated purchases because items came back.
- D) Most items returned, and students buy things sometimes.

254. Which sentence uses apostrophes correctly?

- A) The two coaches' schedules were posted beside the players' lockers.
- B) The two coach's schedules were posted beside the player's lockers.
- C) The two coaches schedules were posted beside the players lockers.
- D) The two coaches's schedules were posted beside the players's lockers.

255. Which sentence is punctuated correctly?

- A) Nina said "I will bring the poster tomorrow."
- B) Nina said "I will bring the poster tomorrow".
- C) Nina said, "I will bring the poster tomorrow."
- D) Nina, said, "I will bring the poster tomorrow."

256. Which sentence uses capitalization correctly?

- A) Professor Lee teaches Biology at westfield college.
- B) professor Lee teaches Biology at Westfield College.
- C) Professor Lee teaches biology at Westfield College.
- D) Professor lee teaches biology at Westfield college.

257. Which sentence uses capitalization correctly?

- A) Our Family crossed the Mississippi river during our Summer trip.
- B) Our family crossed the Mississippi River during our Summer Trip.
- C) Our family crossed the mississippi River during our summer trip.
- D) Our family crossed the Mississippi River during our summer trip.

258. Which word is spelled correctly?

- A) accomodate
- B) accommodate
- C) accomodate
- D) accommadate

259. Which sentence contains a misspelled word?

- A) The committee reached a decision.
- B) The route was clearly visible.
- C) Her explanation was thorough.
- D) The package arrived seperately.

260. Which sentence contains no spelling errors?

- A) The audience aplauded the remarkable performance.
- B) The audience applauded the remarkable performance.
- C) The audience applauded the remarkible performance.
- D) The audiance applauded the remarkable performance.

261. Which sentence uses pronouns correctly?

- A) Between you and me, the plan needs revision.
- B) Between you and I, the plan needs revision.
- C) Between yourself and I, the plan needs revision.
- D) Between I and you, the plan needs revision.

262. Which sentence is complete and correctly formed?

- A) The rain stopped before noon, allowing the game to begin.
- B) Although the game beginning after lunch.
- C) Because the rain stopped before noon.
- D) While waiting for the field to dry.

263. Which sentence uses parallel structure correctly?

- A) The club enjoys hiking, to swim, and biking.
- B) The club enjoys hiking, swimming, and biking.
- C) The club enjoys hiking, to swim, and to bike.
- D) The club enjoys to hike, swimming, and to bike.

264. Which sentence uses the most precise word?

- A) The scientist looked at the sample.
- B) The scientist examined the sample.
- C) The scientist did the sample.
- D) The scientist had the sample.

265. Choose the best transition: The trail was muddy after the storm. ____, the hikers continued carefully.

- A) For example
- B) Meanwhile
- C) Similarly
- D) Nevertheless

Editing Passage: A Student Repair Shelf

(1) At North Ridge School, students sometimes discarded binders, folders, and small classroom tools after minor damage. (2) The environmental club proposed a repair shelf where usable items could be fixed instead of thrown away. (3) Volunteers stocked the shelf with tape, replacement rings, labels, and simple instructions. (4) During lunch twice a week, club members helped students decide whether an item could be repaired safely. (5) The shelf soon became popular, but the club noticed that some students left unfinished projects behind. (6) In response, volunteers added labeled trays for work that needed another day. (7) This small change kept the area organized and made it easier for students to return to their projects. (8) The project showed that reducing waste sometimes begins with making repair more convenient.

266. Which sentence would be the best conclusion for the passage?

- A) Some folders are available in many colors.
- B) Lunch periods can feel short on busy days.
- C) The club plans to continue the shelf and teach more students simple repair skills.
- D) The school building has several storage closets.

267. Where would the following sentence best be placed? “The goal was not to repair every object, but to save items that still had useful life.”

- A) After sentence 1
- B) After sentence 5
- C) After sentence 7
- D) After sentence 2

268. Which revision of sentence 5 is clearest?

- A) The shelf became popular because projects were unfinished behind.
- B) The shelf soon became popular, some students left unfinished projects behind.
- C) Popular was the shelf, but unfinished projects behind.
- D) The shelf soon became popular; however, some students left unfinished projects behind.

269. Which sentence uses a colon correctly?

- A) The supplies include: paper, tape, and markers.
- B) Please: bring paper, tape, and markers.
- C) Please bring three supplies: paper, tape, and markers.
- D) The supplies: are paper, tape, and markers.

270. Which sentence is punctuated correctly?

- A) After the final bell rang the students left.
- B) After the final bell rang, the students left.
- C) After, the final bell rang the students left.
- D) After the final bell, rang the students left.

271. Which sentence correctly joins two independent clauses?

- A) The rain ended; the field remained muddy.
- B) The rain ended, the field remained muddy.
- C) The rain ended the field; remained muddy.
- D) The rain; ended, the field remained muddy.

272. Which sentence uses capitalization correctly?

- A) We will visit Yellowstone National park in july.
- B) We will visit yellowstone national park in July.
- C) We will visit Yellowstone National Park in July.
- D) We will visit yellowstone National Park in july.

273. Which sentence uses capitalization correctly?

- A) My aunt speaks Spanish and studies world history.
- B) My aunt speaks spanish and studies world History.
- C) My Aunt speaks spanish and studies World History.
- D) My Aunt speaks Spanish and studies world history.

274. Which word is spelled correctly?

- A) privelege
- B) privilege
- C) priviledge
- D) privilage

275. Which sentence contains a misspelled word?

- A) The mechanic inspected the equipment.
- B) The hikers reached their destinaton before dark.
- C) The students maintained separate folders.
- D) The speaker emphasized the main point.

276. Which sentence has correct subject-verb agreement?

- A) Neither of the answers are seeming correct.
- B) Neither of the answers seem correct.
- C) Neither of the answer seem correct.
- D) Neither of the answers seems correct.

277. Which sentence uses verb tense correctly?

- A) Yesterday, Mia walks to the store and bought bread.
- B) Yesterday, Mia walked to the store and bought bread.
- C) Yesterday, Mia had walk to the store and buys bread.
- D) Yesterday, Mia walking to the store and bought bread.

278. Which sentence places the modifier clearly?

- A) Covered in frost, the window was photographed by Liam.
- B) Covered in frost, Liam photographed the window.
- C) Liam photographed the window covered in frost.
- D) Photographing, the frost covered Liam's window.

279. Choose the sentence with the clearest pronoun reference.

- A) Maya was nervous when she spoke to Tessa.
- B) When Maya spoke to Tessa, this made her nervous.
- C) She was nervous when Maya spoke to Tessa.
- D) When Maya spoke to Tessa, she was nervous.

280. Which sentence is most concise without changing the meaning?

- A) The reason is because the bus was late.
- B) Heavy traffic made the bus late.
- C) The bus was late due to the fact that traffic was heavy.
- D) The bus was late on account of the fact of heavy traffic.

281. Choose the best word to complete the sentence: The new schedule will ____ how early the team must arrive.

- A) affect
- B) except
- C) effect
- D) accept

Editing Passage: A Better Lost-and-Found System

(1) The student council noticed that the school's lost-and-found table became crowded within a few weeks of each semester. (2) Jackets covered smaller objects, and students often searched through the same pile several times. (3) The council proposed replacing the single table with labeled bins for clothing, school supplies, water bottles, and other items. (4) Members also created a weekly display of newly found objects near the cafeteria entrance. (5) After one month, fewer items remained unclaimed. (6) Students said the categories made searching faster, while the display reminded them to check for missing belongings. (7) The council decided to keep both changes for the rest of the year. (8) The project required little money, but a clearer system made the existing lost-and-found much easier to use.

282. Which sentence would best follow sentence 3?

- A) Some students prefer backpacks with several pockets.
- B) Council elections take place each spring.
- C) The cafeteria serves lunch in three periods.
- D) Each bin received a large sign that could be read from several feet away.

283. Which revision best combines sentences 5 and 6?

- A) After one month fewer items, remaining unclaimed, students said categories and displays.
- B) After one month, fewer items remained unclaimed because the categories made searching faster and the display reminded students to check for missing belongings.
- C) Fewer items remained unclaimed and students said things about the system after one month.
- D) The categories made searching faster, after one month, and fewer items, while reminders.

284. Which sentence uses quotation marks and punctuation correctly?

- A) Maya asked "Are we leaving now"?
- B) Maya asked, "Are we leaving now".
- C) Maya asked "Are we leaving now?"
- D) Maya asked, "Are we leaving now?"

285. Which sentence uses commas correctly?

- A) Before dinner we finished, the puzzle.
- B) Before dinner, we finished the puzzle.
- C) Before dinner we, finished the puzzle.
- D) Before, dinner we finished the puzzle.

286. Which sentence uses capitalization correctly?

- A) professor Elena Ruiz will speak on Monday.
- B) Professor Elena Ruiz will speak on Monday.
- C) Professor Elena Ruiz will speak on monday.
- D) Professor elena Ruiz will speak on Monday.

287. Which sentence uses capitalization correctly?

- A) The club visited the Pacific Ocean during spring break.
- B) The Club visited the pacific ocean during Spring Break.
- C) The club visited the Pacific ocean during spring break.
- D) The club visited the pacific Ocean during spring break.

288. Which word is spelled correctly?

- A) occurence
- B) occurance
- C) occurrence
- D) occurrence

289. Which sentence contains a misspelled word?

- A) The committee reached a unanimous decision.
- B) The athlete showed remarkable endurance.
- C) The schedule is available immediately.
- D) The museum displayed an anciet manuscript.

290. Which sentence contains no spelling errors?

- A) The audience applauded the performance enthusiastically.
- B) The audience aplauded the performance enthusiastically.
- C) The audience applauded the performance enthusiastically.
- D) The audiance applauded the performance enthusiastically.

291. Which sentence uses pronoun case correctly?

- A) The coach spoke to Jordan and I.
- B) The coach spoke to I and Jordan.
- C) The coach spoke to Jordan and me.
- D) The coach spoke to me and he.

292. Which sentence avoids a run-on sentence?

- A) The lights dimmed the audience became quiet.
- B) The lights dimmed, and the audience became quiet.
- C) The lights dimmed, the audience became quiet.
- D) The lights dimmed and, the audience became quiet.

293. Which sentence has correct pronoun agreement?

- A) Every student should bring his or her signed form to the office.
- B) Every students should bring his signed form to the office.
- C) Every student should brings his or her signed form to the office.
- D) Every student should bring their signed form to the office.

294. Choose the best word to complete the sentence: The weather forecast may ____ our decision to hold the picnic outdoors.

- A) elude
- B) altar
- C) allude
- D) alter

295. Which sentence expresses the relationship most logically?

- A) Although the battery was fully charged, the device operated for several hours.
- B) Unless the battery was fully charged, therefore the device operated for several hours.
- C) Because the battery was fully charged, the device operated for several hours.
- D) The battery was fully charged; nevertheless, that is why the device operated for several hours.

296. Which sentence is most appropriate for formal written English?

- A) The results surprised all of the participants.
- B) The results were, like, surprising to everyone.
- C) The results were kind of surprising to everybody.
- D) Everybody was super surprised by the results.

Editing Passage: Turning an Empty Corner into a Reading Space

(1) A narrow corner beside the school library's windows had long held two unused tables and several empty boxes. (2) The library committee decided to turn the area into a small reading space. (3) Students measured the corner and found chairs that could fit without blocking the walkway. (4) They also moved a low shelf there and filled it with short-story collections and magazines. (5) A student suggested adding a large television for video games. (6) The committee instead chose two floor lamps because the corner became dim late in the afternoon. (7) After the space opened, students began using it during lunch and before after-school activities. (8) The project showed that a useful improvement does not always require a large room or an expensive renovation.

297. Which sentence should be removed to improve the passage's unity?

- A) Sentence 5
- B) Sentence 4
- C) Sentence 7
- D) Sentence 2

298. Which sentence would best serve as an additional conclusion after sentence 8?

- A) Some magazines are published every month.
- B) The school building also contains a gymnasium and several science rooms.
- C) Careful planning had turned an overlooked corner into a place students actually wanted to use.
- D) The committee meets on Tuesdays after school.

Answer Key

Question numbers continue across all five sections.

Verbal Skills

1. C	2. A	3. B	4. C	5. D	6. D	7. B	8. C
9. A	10. B	11. A	12. A	13. D	14. B	15. D	16. A
17. C	18. B	19. D	20. D	21. A	22. B	23. D	24. D
25. A	26. D	27. C	28. A	29. C	30. A	31. A	32. B
33. B	34. D	35. B	36. A	37. D	38. B	39. C	40. C
41. B	42. D	43. C	44. C	45. D	46. C	47. B	48. D
49. C	50. A	51. B	52. A	53. C	54. B	55. A	56. B
57. A	58. D	59. C	60. C				

Quantitative Skills

61. C	62. D	63. D	64. A	65. B	66. A	67. A	68. B
69. C	70. A	71. B	72. B	73. A	74. B	75. C	76. D
77. B	78. C	79. D	80. A	81. B	82. D	83. B	84. D
85. D	86. B	87. A	88. D	89. A	90. A	91. C	92. A
93. C	94. D	95. C	96. C	97. A	98. C	99. D	100. C
101. B	102. A	103. D	104. B	105. B	106. C	107. A	108. D
109. B	110. C	111. C	112. D				

Reading Comprehension

113. C	114. D	115. D	116. C	117. B	118. D	119. A	120. A
121. D	122. B	123. A	124. D	125. C	126. B	127. D	128. A
129. B	130. A	131. D	132. A	133. D	134. B	135. C	136. A
137. B	138. B	139. C	140. B	141. C	142. A	143. B	144. A
145. C	146. B	147. C	148. B	149. D	150. C	151. D	152. C
153. B	154. D	155. A	156. D	157. D	158. B	159. B	160. D
161. C	162. D	163. A	164. A	165. C	166. B	167. A	168. A
169. B	170. C	171. A	172. C	173. C	174. A		

Mathematics

175. B	176. A	177. B	178. C	179. B	180. D	181. A	182. B
183. C	184. B	185. D	186. B	187. D	188. A	189. B	190. B
191. A	192. D	193. D	194. C	195. D	196. D	197. B	198. A
199. C	200. A	201. C	202. C	203. D	204. C	205. A	206. D

207. D	208. A	209. B	210. C	211. C	212. B	213. C	214. B
215. A	216. D	217. A	218. C	219. A	220. C	221. D	222. B
223. D	224. C	225. B	226. D	227. A	228. A	229. C	230. A
231. A	232. C	233. B	234. B	235. C	236. D	237. D	238. A

Language

239. C	240. D	241. C	242. C	243. D	244. B	245. A	246. D
247. A	248. C	249. C	250. D	251. A	252. D	253. B	254. A
255. C	256. C	257. D	258. B	259. D	260. B	261. A	262. A
263. B	264. B	265. D	266. C	267. D	268. D	269. C	270. B
271. A	272. C	273. A	274. B	275. B	276. D	277. B	278. A
279. A	280. B	281. A	282. D	283. B	284. D	285. B	286. B
287. A	288. C	289. D	290. C	291. C	292. B	293. A	294. D
295. C	296. A	297. A	298. C				

Detailed Explanations

Verbal Skills

1. HSPT-V8-001 - Answer C: careful

Deliberate can mean carefully considered or intentional, so careful is the closest choice here.

2. HSPT-V8-002 - Answer A: hide

To obscure something is to make it difficult to see or understand, so hide is the closest synonym.

3. HSPT-V8-003 - Answer B: beginner

A novice is someone new to an activity, so beginner is the closest synonym.

4. HSPT-V8-004 - Answer C: sturdy

Fragile means easily damaged, while sturdy means strong and resistant to damage.

5. HSPT-V8-005 - Answer D: rare

Frequent means happening often, while rare means happening infrequently.

6. HSPT-V8-006 - Answer D: selfish

Generous describes willingness to give; selfish describes concern mainly for oneself.

7. HSPT-V8-007 - Answer B: automobile

A rudder controls the direction of a ship, while a steering wheel controls the direction of an automobile.

8. HSPT-V8-008 - Answer C: distance

Scales measure weight, while a ruler measures length or distance.

9. HSPT-V8-009 - Answer A: statue

An author creates a novel, just as a sculptor creates a statue.

10. HSPT-V8-010 - Answer B: evaporate

Orbit, rotate, and revolve describe types of motion; evaporate describes a change of state.

11. HSPT-V8-011 - Answer A: pine

Maple, birch, oak, and pine are all trees.

12. HSPT-V8-012 - Answer A: paragraph

Fraction, decimal, and percent are forms of numerical representation; paragraph is a unit of writing.

13. HSPT-V8-013 - Answer D: science

Art must come first, music second, and science after music, so science must be in the afternoon.

14. HSPT-V8-014 - Answer B: No bronze token is stored in Box C.

All bronze tokens are round, and no round token is in Box C, so no bronze token can be there.

15. HSPT-V8-015 - Answer D: M

The constraints force the order M, J, K, L, so M must be first.

16. HSPT-V8-016 - Answer A: economical

Frugal means careful about spending or using resources, so economical is closest.

17. HSPT-V8-017 - Answer C: reduce

To diminish means to make or become smaller or less.

18. HSPT-V8-018 - Answer B: watchful

Vigilant means alert and watchful for possible problems.

19. HSPT-V8-019 - Answer D: flexible

Rigid means stiff or inflexible, so flexible is its clearest opposite.

20. HSPT-V8-020 - Answer D: optional

Compulsory means required, whereas optional means not required.

21. HSPT-V8-021 - Answer A: permanent

Temporary means lasting for a limited time; permanent means lasting indefinitely.

22. HSPT-V8-022 - Answer B: frog

A larva develops into a butterfly, just as a tadpole develops into a frog.

23. HSPT-V8-023 - Answer D: turtle

Bark is an outer covering of a tree, and a shell is an outer covering of a turtle.

24. HSPT-V8-024 - Answer D: teacher
An apprentice learns from a master, just as a student learns from a teacher.
25. HSPT-V8-025 - Answer A: trumpet
Violin, cello, and viola are bowed string instruments; trumpet is a brass instrument.
26. HSPT-V8-026 - Answer D: silver
Copper, iron, aluminum, and silver are metals.
27. HSPT-V8-027 - Answer C: enormous
Tiny, minute, and miniature describe something very small, while enormous describes something very large.
28. HSPT-V8-028 - Answer A: P
Since S must be immediately before P and R cannot be last, the only valid ordering places P last.
29. HSPT-V8-029 - Answer C: The library is not open.
By contraposition, if availability is required whenever the library is open, unavailable means the library cannot be open.
30. HSPT-V8-030 - Answer A: Every cedar box is wooden.
The first statement directly guarantees that every cedar box belongs to the wooden-box category.
31. HSPT-V8-031 - Answer A: precise
Meticulous means extremely careful and precise about details.
32. HSPT-V8-032 - Answer B: possible
Feasible means possible or practical to accomplish.
33. HSPT-V8-033 - Answer B: honest
Candid means open and honest in expression.
34. HSPT-V8-034 - Answer D: famous
Obscure can mean little known; famous is its clearest opposite in this context.
35. HSPT-V8-035 - Answer B: contract
To expand is to become larger; to contract is to become smaller.
36. HSPT-V8-036 - Answer A: cautious
Reckless means acting without enough care, while cautious means careful about risk.
37. HSPT-V8-037 - Answer D: wall
A page is a component of a book, while a brick is a component of a wall.
38. HSPT-V8-038 - Answer B: bird
Scales cover a fish, while feathers cover a bird.
39. HSPT-V8-039 - Answer C: judge
A navigator characteristically uses a sextant, while a judge characteristically uses a gavel.
40. HSPT-V8-040 - Answer C: fern
Orchid, tulip, and daffodil are flowering plants; a fern reproduces without flowers.
41. HSPT-V8-041 - Answer B: Saturn
Mercury, Venus, Mars, and Saturn are planets.
42. HSPT-V8-042 - Answer D: sphere
Triangle, pentagon, and hexagon are two-dimensional polygons; a sphere is a three-dimensional solid.
43. HSPT-V8-043 - Answer C: W
The constraints force W above X above Z above Y, so W must be on top.
44. HSPT-V8-044 - Answer C: Priya attends the safety briefing.
The rule applies to every robotics-team member, and Priya is explicitly identified as one.
45. HSPT-V8-045 - Answer D: Tuesday
Pottery is offered only Tuesday or Thursday, and Thursday is excluded, so Elena must attend Tuesday.
46. HSPT-V8-046 - Answer C: durable
Resilient describes something able to recover or remain strong, making durable the closest choice.
47. HSPT-V8-047 - Answer B: brief
Concise means expressed in few words without unnecessary detail.
48. HSPT-V8-048 - Answer D: confirm
To verify something is to confirm that it is accurate or true.
49. HSPT-V8-049 - Answer C: scarce
Abundant means plentiful, while scarce means present in limited supply.

50. HSPT-V8-050 - Answer A: restless
Tranquil means calm or peaceful; restless is the clearest opposite.
51. HSPT-V8-051 - Answer B: refuse
To permit is to allow something; to refuse it is to deny permission.
52. HSPT-V8-052 - Answer A: fox
A hive is a dwelling associated with a bee, and a den is a dwelling associated with a fox.
53. HSPT-V8-053 - Answer C: pottery
Wool is material used to make a sweater; clay is material used to make pottery.
54. HSPT-V8-054 - Answer B: rest
Eating responds to hunger, just as resting responds to fatigue.
55. HSPT-V8-055 - Answer A: quartz
Quartz is a mineral, while granite, marble, and limestone are rocks.
56. HSPT-V8-056 - Answer B: metals
Copper, silver, and gold are all metals.
57. HSPT-V8-057 - Answer A: ravine
A ravine, like a canyon, valley, and gorge, is a low landform between higher sides.
58. HSPT-V8-058 - Answer D: S
The order constraints force S before R before N before P before T, so S must finish first.
59. HSPT-V8-059 - Answer C: It is not marked EXPRESS.
By the contrapositive of the stated rule, a package not on Cart 2 cannot be marked EXPRESS.
60. HSPT-V8-060 - Answer C: Room B is closed.
If exactly one of the two rooms is open and Room A is open, Room B must be closed.

Quantitative Skills

61. HSPT-Q8-001 - Answer C: 41

The differences increase by 1: +5, +6, +7, +8, so the next change is +9.

62. HSPT-Q8-002 - Answer D: 8

Each term is half the previous term, so $16 \div 2 = 8$.

63. HSPT-Q8-003 - Answer D: 12

The pattern alternates +4 and -1, so $13 - 1 = 12$.

64. HSPT-Q8-004 - Answer A: 81

Each term is multiplied by 3, so $27 \times 3 = 81$.

65. HSPT-Q8-005 - Answer B: 7/20

$35\% = 35/100$, which simplifies to 7/20.

66. HSPT-Q8-006 - Answer A: 49

Multiply 28 by 7/4 to get 49.

67. HSPT-Q8-007 - Answer A: 30%

The increase is 15, and $15 \div 50 = 0.30$, or 30%.

68. HSPT-Q8-008 - Answer B: The two quantities are equal.

5/8 of 64 is 40, equal to Quantity B.

69. HSPT-Q8-009 - Answer C: Quantity B is greater.

Solving gives $x = 6$, so Quantity B is greater.

70. HSPT-Q8-010 - Answer A: The relationship cannot be determined from the information given.

If $a = 7$ the quantities are equal, while larger values make Quantity A greater.

71. HSPT-Q8-011 - Answer B: Quantity A is greater.

The square's area is 49 cm^2 , which is greater than 48 cm^2 .

72. HSPT-Q8-012 - Answer B: Quantity B is greater.

The triangle's perimeter is 27 cm, which is less than 28 cm.

73. HSPT-Q8-013 - Answer A: The two quantities are equal.

The cube's volume is $3^3 = 27 \text{ cm}^3$.

74. HSPT-Q8-014 - Answer B: 95

Each term is twice the previous term plus 1, so $47 \times 2 + 1 = 95$.

75. HSPT-Q8-015 - Answer C: 55

The differences are -9, -8, -7, -6, so the next difference is -5.

76. HSPT-Q8-016 - Answer D: 13.5

The pattern alternates $\times 3$ and $+2$, so $27 \div 2 = 13.5$.

77. HSPT-Q8-017 - Answer B: 36

Undo the operations: $16 - 7 = 9$, then $9 \times 4 = 36$.

78. HSPT-Q8-018 - Answer C: 11

Add 9 to obtain $4n = 44$, then divide by 4 to get $n = 11$.

79. HSPT-Q8-019 - Answer D: 56

Since 35 is 7 times 5, multiply 8 by 7 to get 56.

80. HSPT-Q8-020 - Answer A: 3/4

$3/4 = 0.75$, which is greater than 0.72, 0.71, and 0.70.

81. HSPT-Q8-021 - Answer B: The two quantities are equal.

Quantity A is $8 + 5 = 13$ and Quantity B is $9 + 4 = 13$.

82. HSPT-Q8-022 - Answer D: Quantity B is greater.

Quantity A is 30, while Quantity B is 32.

83. HSPT-Q8-023 - Answer B: Quantity B is greater.

Because $n < 8$, $n + 4 < 12$, so Quantity B must be greater.

84. HSPT-Q8-024 - Answer D: Quantity A is greater.

The diameter is twice the radius, so Quantity A is 16 cm, greater than 15 cm.

85. HSPT-Q8-025 - Answer D: The two quantities are equal.
The triangle's area is $\frac{1}{2} \times 10 \times 6 = 30 \text{ cm}^2$.
86. HSPT-Q8-026 - Answer B: Quantity A is greater.
The third angle is $180^\circ - 35^\circ - 75^\circ = 70^\circ$, which is greater than 65° .
87. HSPT-Q8-027 - Answer A: 46
The increases are 3, 6, 9, and 12; the next increase is 15, giving 46.
88. HSPT-Q8-028 - Answer D: 10
Each term is half the previous term, so $20 \div 2 = 10$.
89. HSPT-Q8-029 - Answer A: 9
Two interwoven sequences increase by 2: 3, 5, 7, 9 and 8, 10, 12.
90. HSPT-Q8-030 - Answer A: 16.2
Each term is multiplied by 3, so $5.4 \times 3 = 16.2$.
91. HSPT-Q8-031 - Answer C: \$45
Twenty-five percent of \$60 is \$15, so the sale price is $\$60 - \$15 = \$45$.
92. HSPT-Q8-032 - Answer A: $\frac{3}{8}$
 $0.375 = \frac{375}{1000}$, which simplifies to $\frac{3}{8}$.
93. HSPT-Q8-033 - Answer C: \$15
Each notebook costs \$3, so 5 notebooks cost $5 \times \$3 = \15 .
94. HSPT-Q8-034 - Answer D: The two quantities are equal.
Both quantities equal 9.
95. HSPT-Q8-035 - Answer C: Quantity A is greater.
 $\frac{5}{6}$ of 42 is 35, which is greater than 34.
96. HSPT-Q8-036 - Answer C: The relationship cannot be determined from the information given.
If $x = 7$ the quantities are equal, but if $x > 7$ Quantity A is greater.
97. HSPT-Q8-037 - Answer A: Quantity B is greater.
The square's perimeter is $4 \times 7 = 28 \text{ cm}$, which is less than 30 cm.
98. HSPT-Q8-038 - Answer C: The two quantities are equal.
The prism's volume is $2 \times 3 \times 5 = 30 \text{ cm}^3$.
99. HSPT-Q8-039 - Answer D: Quantity B is greater.
The supplementary angle is $180^\circ - 112^\circ = 68^\circ$, so Quantity B is greater.
100. HSPT-Q8-040 - Answer C: 67
The operations alternate multiplying by 3 and adding 1; $66 + 1 = 67$.
101. HSPT-Q8-041 - Answer B: 1
Each term is divided by 3, so $3 \div 3 = 1$.
102. HSPT-Q8-042 - Answer A: 68
The pattern alternates $+2$ and $\times 2$; after 34, multiply by 2 to get 68.
103. HSPT-Q8-043 - Answer D: 60
If 120% of the original is 72, then the original is $72 \div 1.2 = 60$.
104. HSPT-Q8-044 - Answer B: 6
Dividing by $\frac{1}{8}$ means multiplying by 8, so $\frac{3}{4} \times 8 = 6$.
105. HSPT-Q8-045 - Answer B: 12
There are 8 ratio parts total; $32 \div 8 = 4$ per part, and $3 \times 4 = 12$.
106. HSPT-Q8-046 - Answer C: 42
 $42 = 5 \times 8 + 2$, so its remainder when divided by 5 is 2.
107. HSPT-Q8-047 - Answer A: The two quantities are equal.
 $7^2 - 13 = 49 - 13 = 36$, and $6^2 = 36$.
108. HSPT-Q8-048 - Answer D: Quantity B is greater.
Because y is strictly less than 12, Quantity B must be greater.
109. HSPT-Q8-049 - Answer B: The relationship cannot be determined from the information given.
For k below 6, Quantity B is greater; at 6 they are equal; above 6, Quantity A is greater.
110. HSPT-Q8-050 - Answer C: Quantity A is greater.
The circle's area is $3.14 \times 4^2 = 50.24 \text{ cm}^2$, which is greater than 50 cm^2 .

111. HSPT-Q8-051 - Answer C: Quantity B is greater.

The hypotenuse is 13 cm because $5^2 + 12^2 = 13^2$, so 14 cm is greater.

112. HSPT-Q8-052 - Answer D: The two quantities are equal.

The parallelogram's area is $\text{base} \times \text{height} = 9 \times 4 = 36 \text{ cm}^2$.

Reading Comprehension

113. HSPT-R8-001 - Answer C: She wants the repair to disappear.

Tessa assumes exact matches would make the restored area look as though it had never been damaged.

114. HSPT-R8-002 - Answer D: They preserve evidence that the sign has a history of damage and repair.

Ms. Ortiz believes a visible but respectful repair acknowledges the sign's past rather than hiding it.

115. HSPT-R8-003 - Answer D: fit appropriately

The new tiles fit the design even though their shades differed from the original ones.

116. HSPT-R8-004 - Answer C: She learns that a repair can respect history without perfectly copying the original.

Tessa moves from wanting an invisible repair to appreciating a visible repair that preserves the sign's story.

117. HSPT-R8-005 - Answer B: Preserving something can include acknowledging the changes it has undergone.

The restored mosaic remains useful and beautiful while visibly showing part of its history.

118. HSPT-R8-006 - Answer D: Temperature changes

The passage explains that bridge materials change size slightly as temperatures rise and fall.

119. HSPT-R8-007 - Answer A: structural force or strain

Here stress refers to physical strain placed on bridge materials when movement is restricted.

120. HSPT-R8-008 - Answer A: More material can undergo temperature-related expansion and contraction.

The passage states that required movement depends partly on bridge length.

121. HSPT-R8-009 - Answer D: It explains a physical problem, describes an engineering solution, then discusses design and maintenance factors.

The passage moves from thermal movement to expansion joints and then to factors affecting their design and upkeep.

122. HSPT-R8-010 - Answer B: They may become blocked or worn.

The passage mentions dirt, ice, and worn parts as problems that can reduce joint performance.

123. HSPT-R8-011 - Answer A: Archives are useful when materials are preserved, organized, and given enough context to support research.

The passage emphasizes context, organization, description, and preservation as the qualities that make an archive useful.

124. HSPT-R8-012 - Answer D: tools that help researchers locate and understand materials

The phrase appears with descriptions and labels that connect items to useful information.

125. HSPT-R8-013 - Answer C: It can provide evidence about businesses, events, or daily life.

The passage explains that seemingly ordinary records can help reveal larger historical patterns.

126. HSPT-R8-014 - Answer B: analytical and explanatory

The author calmly explains several factors that make archives useful without arguing emotionally.

127. HSPT-R8-015 - Answer D: Formats and storage systems can become outdated.

The passage specifically identifies changing formats and storage systems as digital preservation challenges.

128. HSPT-R8-016 - Answer A: Good organization can increase the research value of ordinary materials.

The passage explains that context, descriptions, and finding aids help researchers use even ordinary records effectively.

129. HSPT-R8-017 - Answer B: He plays one section too quickly.

Jonah explicitly says that he keeps rushing a particular section.

130. HSPT-R8-018 - Answer A: She believes a complete run would feel more useful and satisfying.

Mara wants a flawless full performance and initially sees the short repetition as less worthwhile.

131. HSPT-R8-019 - Answer D: It reveals a hidden problem and changes how the students rehearse.

Slowing the passage exposes Mara's early entrance and leads both students to practice more effectively.

132. HSPT-R8-020 - Answer A: Careful attention to a weak section can improve the larger performance.

The focused practice improves both the troublesome measures and the complete duet.

133. HSPT-R8-021 - Answer D: well practiced

The opening has been practiced repeatedly and is already refined, so polished means well practiced.

134. HSPT-R8-022 - Answer B: Gravity

The passage states that gravity can pull growing droplets toward the beetle's mouth.

135. HSPT-R8-023 - Answer C: To explain a beetle's water-collection strategy and why it interests engineers

The passage explains the biological mechanism and its potential value as inspiration for engineered surfaces.

136. HSPT-R8-024 - Answer A: Engineered designs must also function effectively and practically.

The passage emphasizes realistic performance, durability, useful water collection, and cost.

137. HSPT-R8-025 - Answer B: It describes a natural process, explains the principle behind it, and then discusses possible engineering applications.
The passage moves from beetle behavior to surface science and then to biomimetic design possibilities.
138. HSPT-R8-026 - Answer B: Form and combine
The passage directly states that microscopic surface features help droplets form and combine.
139. HSPT-R8-027 - Answer C: Seed banks require active management to preserve useful plant diversity for future research and use.
The passage emphasizes documentation, storage, testing, renewal, and scientific use.
140. HSPT-R8-028 - Answer B: To reduce moisture and help extend storage life
The passage links drying and low-temperature storage with slowing biological processes and extending storage life.
141. HSPT-R8-029 - Answer C: The renewed sample should continue to represent the original variety.
The passage explicitly warns that regeneration must preserve the characteristics represented by the original sample.
142. HSPT-R8-030 - Answer A: To present a familiar comparison for one protective function of seed banks
The comparison helps readers understand how preserved seeds can provide options after losses.
143. HSPT-R8-031 - Answer B: Seed banks support research and breeding, and their usefulness depends on active preservation practices.
The ending broadens the role of seed banks and emphasizes the management required to sustain their value.
144. HSPT-R8-032 - Answer A: Their germination rate
Workers test how many stored seeds still germinate to monitor viability.
145. HSPT-R8-033 - Answer C: Inside a wooden crate
Avery notices the lantern through a gap in the wooden crate during a scene-change rehearsal.
146. HSPT-R8-034 - Answer B: People assumed the temporary arrangement was obvious.
The crew chief explains that no one mentioned the arrangement because everyone assumed it was obvious.
147. HSPT-R8-035 - Answer C: A communication problem between people using different systems
The lantern is not truly lost; Avery simply lacks information about the crew's temporary storage system.
148. HSPT-R8-036 - Answer B: Clear communication can be as important as careful organization.
Avery's labeled system fails when location changes are not communicated, so she improves both labeling and communication.
149. HSPT-R8-037 - Answer D: was supposed to be located
The sentence concerns the lantern's proper location, so belonged means where it was supposed to be.
150. HSPT-R8-038 - Answer C: Construction has closed the bridge.
The passage states that construction closed the bridge, forcing the club onto a side trail.
151. HSPT-R8-039 - Answer D: They encourage students to make intentional visual choices.
His questions direct attention to framing, viewpoint, and overlooked details rather than supplying subjects.
152. HSPT-R8-040 - Answer C: Careful observation can make ordinary surroundings visually interesting.
Students find strong subjects along an initially ordinary route by paying closer attention.
153. HSPT-R8-041 - Answer B: It moves from disappointment about a detour to discovery and reflection.
The students first resist the new route, then notice details, and finally understand the value of deliberate observation.
154. HSPT-R8-042 - Answer D: To contrast quantity with more thoughtful decision-making
Lila takes fewer pictures because she spends more time considering what each image should contain.
155. HSPT-R8-043 - Answer A: To allow limited movement between bridge sections
Expansion joints provide designed spaces or flexible connections that permit controlled movement.
156. HSPT-R8-044 - Answer D: Expansion joints help bridges manage temperature-related movement and must be designed and maintained carefully.
The passage explains why joints are needed, what affects their design, and why maintenance matters.
157. HSPT-R8-045 - Answer D: Their expected movement may differ because of material, length, and climate.
The passage identifies materials, length, and temperature range as factors affecting expected expansion.
158. HSPT-R8-046 - Answer B: moved from one place to another
The passage means that stress can shift from a faulty joint to other bridge components.
159. HSPT-R8-047 - Answer B: Controlled flexibility can help a structure handle predictable movement.
The final paragraph presents expansion joints as an example of durability achieved through controlled movement.

160. HSPT-R8-048 - Answer D: Shade structures could cost more than expected.
The passage states that they acknowledge the concern that additional structures could cost more than expected.
161. HSPT-R8-049 - Answer C: She believes their research already makes the proposal convincing.
Nadia initially assumes their collected measurements, survey results, and photographs are sufficient to support the proposal.
162. HSPT-R8-050 - Answer D: It introduces the need to consider opposing viewpoints.
The question shifts their preparation from repeating arguments to anticipating objections, which drives the story's main change.
163. HSPT-R8-051 - Answer A: recognizing
They recognize the cost concern directly in their revised proposal rather than pretending it does not exist.
164. HSPT-R8-052 - Answer A: A problem in preparation leads to a revised strategy that succeeds during the showcase.
The narrative moves from Nadia's initial approach through revision to a successful response at the showcase.
165. HSPT-R8-053 - Answer C: At night
The passage explains that some desert plants open their stomata primarily at night when conditions are less drying.
166. HSPT-R8-054 - Answer B: Desert plants use several coordinated adaptations to reduce water loss while carrying out necessary functions.
The passage describes multiple structural and physiological features that work together to conserve limited water.
167. HSPT-R8-055 - Answer A: Nighttime conditions are generally cooler and may reduce evaporation.
The passage connects nighttime stomatal opening with lower temperatures and less drying air, both of which can limit water loss.
168. HSPT-R8-056 - Answer A: a balance involving competing benefits and costs
Plants need gas exchange but risk losing water, so tradeoff refers to balancing competing needs.
169. HSPT-R8-057 - Answer B: To show that water conservation can depend on multiple features working together
The final paragraph emphasizes that survival results from coordinated features rather than one single trait.
170. HSPT-R8-058 - Answer C: Measurements, sketches, and reminders
The passage describes field notebooks containing measurements, sketches, abbreviations, weather notes, and reminders.
171. HSPT-R8-059 - Answer A: They may indicate that the area was difficult to survey.
The passage specifically suggests repeated visits can be evidence that the terrain posed surveying difficulties.
172. HSPT-R8-060 - Answer C: leave out
In context, a surveyor may fail to include information that seemed obvious, so omit means leave out.
173. HSPT-R8-061 - Answer C: They can reveal the decisions and revisions behind a finished map.
The passage emphasizes that notes preserve evidence of observation, uncertainty, checking, and revision behind the final product.
174. HSPT-R8-062 - Answer A: It highlights the difference between a working process and its polished result.
The contrast helps show how rough notes expose the process and uncertainty that a finished map may conceal.

Mathematics

175. HSPT-M8-001 - Answer B: 19

Division and multiplication come first: $24 \div 6 = 4$ and $3 \times 5 = 15$, so the total is 19.

176. HSPT-M8-002 - Answer A: $9/20$

The decimal 0.45 equals $45/100$, which simplifies to $9/20$.

177. HSPT-M8-003 - Answer B: 13

Absolute values are 9 and 4, so their sum is 13.

178. HSPT-M8-004 - Answer C: 49

$49 = 7 \times 7$, so it has factors other than 1 and itself.

179. HSPT-M8-005 - Answer B: 7

Add 7 to get 35, then divide by 5 to get $x = 7$.

180. HSPT-M8-006 - Answer D: $4n + 3$

Four times n is $4n$, and three more gives $4n + 3$.

181. HSPT-M8-007 - Answer A: 13

Substitute $x = 4$: $y = 2(4) + 5 = 13$.

182. HSPT-M8-008 - Answer B: $C = 12 + 6h$

The total combines the fixed \$12 fee with \$6 multiplied by h .

183. HSPT-M8-009 - Answer C: 65°

Triangle angles total 180° , so $x = 180 - 48 - 67 = 65^\circ$.

184. HSPT-M8-010 - Answer B: 70 cm^2

Area equals length $\times$ width: $14 \times 5 = 70 \text{ cm}^2$.

185. HSPT-M8-011 - Answer D: 37.68 cm

Circumference is $2\pi r = 2 \times 3.14 \times 6 = 37.68 \text{ cm}$.

186. HSPT-M8-012 - Answer B: 3,200

One kilometer equals 1,000 meters, so 3.2 kilometers equals 3,200 meters.

187. HSPT-M8-013 - Answer D: 3 cups

$24 \div 8 = 3$ cups.

188. HSPT-M8-014 - Answer A: 11

The value 11 appears more often than any other value.

189. HSPT-M8-015 - Answer B: $1/3$

There are 12 marbles total and 4 are blue, so $4/12 = 1/3$.

190. HSPT-M8-016 - Answer B: 15

Five numbers with mean 14 total 70; the known values total 55, leaving 15.

191. HSPT-M8-017 - Answer A: 18

The greatest positive integer that divides both 54 and 90 is 18.

192. HSPT-M8-018 - Answer D: -2.04

Among negative numbers, the value closest to zero is greatest, so -2.04 is largest.

193. HSPT-M8-019 - Answer D: 80

19.8 rounds to 20 and 4.1 rounds to 4, giving $20 \times 4 = 80$.

194. HSPT-M8-020 - Answer C: $2x + 15$

Distribute first: $6x + 15 - 4x = 2x + 15$.

195. HSPT-M8-021 - Answer D: 11

Expand to $2x - 6 = x + 5$, then subtract x and add 6 to get $x = 11$.

196. HSPT-M8-022 - Answer D: 70 miles

The rate is $42 \div 3 = 14$ miles per hour, and $14 \times 5 = 70$ miles.

197. HSPT-M8-023 - Answer B: $x \geq 6$

At least 6 means 6 or any greater value, represented by $x \geq 6$.

198. HSPT-M8-024 - Answer A: 15 cm

Using the Pythagorean theorem, $9^2 + 12^2 = 81 + 144 = 225$, and $\sqrt{225} = 15$.

199. HSPT-M8-025 - Answer C: 60°
The exterior angles of any polygon total 360° , and $360^\circ \div 6 = 60^\circ$.
200. HSPT-M8-026 - Answer A: 10 cm
The scale factor is $15 \div 6 = 2.5$, so the larger width is $4 \times 2.5 = 10$ cm.
201. HSPT-M8-027 - Answer C: 1 hour 48 minutes
From 6:47 to 7:47 is one hour, and from 7:47 to 8:35 is 48 minutes.
202. HSPT-M8-028 - Answer C: 88 ounces
Five pounds equals 80 ounces; adding 8 ounces gives 88 ounces.
203. HSPT-M8-029 - Answer D: 63 miles
Multiply the map distance by the scale: $3.5 \times 18 = 63$ miles.
204. HSPT-M8-030 - Answer C: 24
The values are already ordered, and the middle of five values is 24.
205. HSPT-M8-031 - Answer A: $\frac{3}{4}$
Six of the eight sections are not green, so the probability is $\frac{6}{8} = \frac{3}{4}$.
206. HSPT-M8-032 - Answer D: 16
A mean of 14 over five games requires 70 points total; the first four total 54, so 16 more are needed.
207. HSPT-M8-033 - Answer D: 53
53 has no positive factors other than 1 and 53.
208. HSPT-M8-034 - Answer A: 33
Evaluate parentheses and exponent first: $(8 - 5)^2 = 9$; then $3 \times 9 = 27$ and $6 + 27 = 33$.
209. HSPT-M8-035 - Answer B: 2.6
 $\frac{3}{5} = 0.6$, so $2 \frac{3}{5} = 2.6$.
210. HSPT-M8-036 - Answer C: 84
 $0.35 \times 240 = 84$.
211. HSPT-M8-037 - Answer C: 19
Substitute 7 for x: $3(7) - 2 = 21 - 2 = 19$.
212. HSPT-M8-038 - Answer B: 11
Subtract $3x$ from both sides and subtract 7 to get $x = 11$.
213. HSPT-M8-039 - Answer C: $2m + 4$
The fixed fee is 4 and the mileage charge is $2m$, so the total is $2m + 4$.
214. HSPT-M8-040 - Answer B: 7
Set $36 = 5x + 1$; subtract 1 to get $35 = 5x$, so $x = 7$.
215. HSPT-M8-041 - Answer A: 12 units
Length = area $\div$ width = $96 \div 8 = 12$ units.
216. HSPT-M8-042 - Answer D: 65°
Triangle angles total 180° , so $180^\circ - 48^\circ - 67^\circ = 65^\circ$.
217. HSPT-M8-043 - Answer A: 9 inches
The radius is half the diameter, so $18 \div 2 = 9$ inches.
218. HSPT-M8-044 - Answer C: 2,750
One liter equals 1,000 milliliters, so 2.75 liters equals 2,750 milliliters.
219. HSPT-M8-045 - Answer A: 26 m
Multiply the drawing length by the scale: $6.5 \times 4 = 26$ meters.
220. HSPT-M8-046 - Answer C: 8
The range is the greatest value minus the least value: $14 - 6 = 8$.
221. HSPT-M8-047 - Answer D: $\frac{1}{22}$
The probability is $\frac{3}{12} \times \frac{2}{11} = \frac{6}{132} = \frac{1}{22}$.
222. HSPT-M8-048 - Answer B: 20
Five numbers with mean 18 total 90; the known numbers total 70, so the missing number is 20.
223. HSPT-M8-049 - Answer D: 25
25 has the factors 1, 5, and 25, exactly three positive factors.
224. HSPT-M8-050 - Answer C: 8
Because $8^2 = 64$ and $9^2 = 81$, $\sqrt{70}$ lies between 8 and 9 and is closer to 8.

225. HSPT-M8-051 - Answer B: 40%
If $\frac{3}{5}$ is full, $\frac{2}{5}$ is empty, and $\frac{2}{5} = 40\%$.
226. HSPT-M8-052 - Answer D: $-a + 8$
Distribute -2 to get $5a - 6a + 8$, which simplifies to $-a + 8$.
227. HSPT-M8-053 - Answer A: 7
Expand to $3x + 12 = 2x + 19$; subtract $2x$ and 12 to obtain $x = 7$.
228. HSPT-M8-054 - Answer A: $x < 7$
Add 5 to get $2x < 14$, then divide by 2 to obtain $x < 7$.
229. HSPT-M8-055 - Answer C: $g = 18 + 4t$
The tank starts with 18 gallons and gains 4 gallons each minute, so $g = 18 + 4t$.
230. HSPT-M8-056 - Answer A: 136 cm^2
Surface area is $2(8 \times 3 + 8 \times 4 + 3 \times 4) = 2(24 + 32 + 12) = 136 \text{ cm}^2$.
231. HSPT-M8-057 - Answer A: $6\sqrt{2}$
By the Pythagorean theorem, $d^2 = 6^2 + 6^2 = 72$, so $d = 6\sqrt{2}$.
232. HSPT-M8-058 - Answer C: 540°
A pentagon's interior-angle sum is $(5 - 2) \times 180^\circ = 540^\circ$.
233. HSPT-M8-059 - Answer B: $2,640$ yards
 $1.5 \times 1,760 = 2,640$ yards.
234. HSPT-M8-060 - Answer B: 12 cups
Multiply 3 quarts by 4 cups per quart to get 12 cups.
235. HSPT-M8-061 - Answer C: 180
The rate is 3 parts per minute; in 60 minutes the machine produces $3 \times 60 = 180$ parts.
236. HSPT-M8-062 - Answer D: 9
With five ordered values, the median is the middle value, 9 .
237. HSPT-M8-063 - Answer D: $\frac{1}{2}$
There are 5 favorable tiles out of 10 total, so the probability is $\frac{5}{10} = \frac{1}{2}$.
238. HSPT-M8-064 - Answer A: It decreases by 4 points.
The original mean is 84 ; without 100 , the mean is 80 , so it decreases by 4 points.

Language

239. HSPT-L8-001 - Answer C: Before the concert began, the audience grew quiet.
A comma follows the introductory dependent clause before the main clause.
240. HSPT-L8-002 - Answer D: The kit contained three items: rope, tape, and gloves.
A colon correctly introduces a list after a complete clause.
241. HSPT-L8-003 - Answer C: The lights dimmed; the performance began.
A semicolon can join two closely related independent clauses.
242. HSPT-L8-004 - Answer C: We studied French history in March.
The language-derived adjective French and the month March are capitalized; history is not.
243. HSPT-L8-005 - Answer D: The class visited Yellowstone National Park on Monday.
The official park name and Monday are capitalized.
244. HSPT-L8-006 - Answer B: necessary
Necessary is the standard spelling.
245. HSPT-L8-007 - Answer A: The speaker gave a relevent example.
Relevant is misspelled; the correct spelling is relevant.
246. HSPT-L8-008 - Answer D: Each of the players wears a numbered jersey.
The singular subject each requires the singular verb wears.
247. HSPT-L8-009 - Answer A: By noon, the workers had finished the repair.
Had finished is the correct past perfect form.
248. HSPT-L8-010 - Answer C: Walking across the plaza, Diego noticed the fountain.
The introductory phrase correctly modifies Diego, the person who was walking.
249. HSPT-L8-011 - Answer C: The storm may affect the schedule.
Affect is the verb meaning to influence.
250. HSPT-L8-012 - Answer D: Everyone agreed unanimously.
This sentence states the idea clearly without redundant words.
251. HSPT-L8-013 - Answer A: A simple lending system helped students share occasionally needed supplies.
This sentence introduces the passage's main idea and previews the purpose of the station.
252. HSPT-L8-014 - Answer D: These changes made expectations clearer for borrowers.
The sentence connects the new reminders with the improved return behavior described next.
253. HSPT-L8-015 - Answer B: By the end of the month, most items were returned on time, and the station reduced unnecessary purchases even though students still needed to own some supplies.
This revision combines both ideas clearly while preserving the original meaning.
254. HSPT-L8-016 - Answer A: The two coaches' schedules were posted beside the players' lockers.
Plural possessive nouns ending in s take an apostrophe after the s.
255. HSPT-L8-017 - Answer C: Nina said, "I will bring the poster tomorrow."
A comma introduces the direct quotation, and the period belongs inside the closing quotation mark.
256. HSPT-L8-018 - Answer C: Professor Lee teaches biology at Westfield College.
The title before the name and the college's proper name are capitalized; the general subject biology is not.
257. HSPT-L8-019 - Answer D: Our family crossed the Mississippi River during our summer trip.
Mississippi River is a proper geographic name, while family, summer, and trip are common nouns.
258. HSPT-L8-020 - Answer B: accommodate
Accommodate is spelled with double c and double m.
259. HSPT-L8-021 - Answer D: The package arrived seperately.
Separately is the correct spelling; seperately is misspelled.
260. HSPT-L8-022 - Answer B: The audience applauded the remarkable performance.
Audience, applauded, remarkable, and performance are all spelled correctly.
261. HSPT-L8-023 - Answer A: Between you and me, the plan needs revision.
The preposition between requires objective pronouns, so me is correct.
262. HSPT-L8-024 - Answer A: The rain stopped before noon, allowing the game to begin.
This choice contains a complete independent clause and a correctly attached modifying phrase.

263. HSPT-L8-025 - Answer B: The club enjoys hiking, swimming, and biking.
All three activities use the same gerund form, creating parallel structure.
264. HSPT-L8-026 - Answer B: The scientist examined the sample.
Examined precisely describes careful observation of a scientific sample.
265. HSPT-L8-027 - Answer D: Nevertheless
Nevertheless signals the contrast between difficult trail conditions and the hikers' decision to continue.
266. HSPT-L8-028 - Answer C: The club plans to continue the shelf and teach more students simple repair skills.
This conclusion extends the passage's focus on practical repair and waste reduction.
267. HSPT-L8-029 - Answer D: After sentence 2
The sentence directly clarifies the purpose of the repair shelf immediately after the proposal is introduced.
268. HSPT-L8-030 - Answer D: The shelf soon became popular; however, some students left unfinished projects behind.
This revision clearly connects the contrasting ideas and avoids a comma splice.
269. HSPT-L8-031 - Answer C: Please bring three supplies: paper, tape, and markers.
A colon correctly follows a complete clause that introduces a list.
270. HSPT-L8-032 - Answer B: After the final bell rang, the students left.
A comma follows the introductory dependent clause.
271. HSPT-L8-033 - Answer A: The rain ended; the field remained muddy.
A semicolon can correctly join two closely related independent clauses.
272. HSPT-L8-034 - Answer C: We will visit Yellowstone National Park in July.
The official park name and the month July are proper nouns and must be capitalized.
273. HSPT-L8-035 - Answer A: My aunt speaks Spanish and studies world history.
Language names are capitalized, while a general school subject such as world history is lowercase.
274. HSPT-L8-036 - Answer B: privilege
Privilege is the standard spelling.
275. HSPT-L8-037 - Answer B: The hikers reached their destination before dark.
Destination is the correct spelling; destination is missing an i.
276. HSPT-L8-038 - Answer D: Neither of the answers seems correct.
The subject neither is singular, so it takes the singular verb seems.
277. HSPT-L8-039 - Answer B: Yesterday, Mia walked to the store and bought bread.
Both completed actions occurred yesterday, so both verbs correctly use the simple past tense.
278. HSPT-L8-040 - Answer A: Covered in frost, the window was photographed by Liam.
The introductory phrase Covered in frost clearly modifies the window, not Liam.
279. HSPT-L8-041 - Answer A: Maya was nervous when she spoke to Tessa.
Naming Maya as the subject makes it clear who was nervous.
280. HSPT-L8-042 - Answer B: Heavy traffic made the bus late.
This version states the cause directly without redundant or wordy phrasing.
281. HSPT-L8-043 - Answer A: affect
Affect is the verb meaning to influence something.
282. HSPT-L8-044 - Answer D: Each bin received a large sign that could be read from several feet away.
This sentence develops the new organization system introduced in sentence 3.
283. HSPT-L8-045 - Answer B: After one month, fewer items remained unclaimed because the categories made searching faster and the display reminded students to check for missing belongings.
The revision clearly connects the result with the two reasons students gave for the improvement.
284. HSPT-L8-046 - Answer D: Maya asked, "Are we leaving now?"
A direct quotation is introduced with a comma, and the question mark belongs inside the closing quotation mark.
285. HSPT-L8-047 - Answer B: Before dinner, we finished the puzzle.
The introductory phrase Before dinner is correctly followed by a comma.
286. HSPT-L8-048 - Answer B: Professor Elena Ruiz will speak on Monday.
A title used directly before a name, the person's name, and the day Monday are capitalized.
287. HSPT-L8-049 - Answer A: The club visited the Pacific Ocean during spring break.
Pacific Ocean is a proper geographic name; club and spring break are common terms here.

288. HSPT-L8-050 - Answer C: occurrence
Occurrence is spelled with double c and double r.
289. HSPT-L8-051 - Answer D: The museum displayed an anciet manuscript.
Ancient is the correct spelling; anciet is missing the second n.
290. HSPT-L8-052 - Answer C: The audience applauded the performance enthusiastically.
Audience, applauded, performance, and enthusiastically are all spelled correctly.
291. HSPT-L8-053 - Answer C: The coach spoke to Jordan and me.
The pronoun is the object of the preposition to, so the objective form me is required.
292. HSPT-L8-054 - Answer B: The lights dimmed, and the audience became quiet.
Two independent clauses are correctly joined by a comma and the coordinating conjunction and.
293. HSPT-L8-055 - Answer A: Every student should bring his or her signed form to the office.
The singular antecedent every student is matched with a singular pronoun phrase and the correct verb form.
294. HSPT-L8-056 - Answer D: alter
Alter means to change, which fits the meaning of the sentence.
295. HSPT-L8-057 - Answer C: Because the battery was fully charged, the device operated for several hours.
Because clearly expresses the cause-and-effect relationship between a charged battery and extended operation.
296. HSPT-L8-058 - Answer A: The results surprised all of the participants.
This sentence uses precise, concise wording appropriate for formal written English.
297. HSPT-L8-059 - Answer A: Sentence 5
The video-game suggestion is not developed and distracts from the passage's focus on creating a practical reading space.
298. HSPT-L8-060 - Answer C: Careful planning had turned an overlooked corner into a place students actually wanted to use.
This conclusion reinforces the passage's central idea that thoughtful planning transformed an underused space.